

The tree of standard Young tableaux: automorphisms and the rank census

On Problems 2 and 3 of D. R. Worley’s notebook of open problems
(arXiv:2509.25446)

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Provenance. Prepared by an AI research session (Claude) directed and reviewed by [NAMES]. All statements are labelled PROVED, COMPUTATIONALLY VERIFIED (with range), or CONJECTURAL; code and data at [REPO URL].

Abstract. Worley’s Problem 3 concerns the tree SYT of all standard Young tableaux ordered by containment and makes two conjectures: (A) every automorphism of the tree is a composition of the “partial transposes” τ_T at vertices T of symmetric shape, and (B) the rank census of the subtree below a vertex determines its shape up to transpose. We show that (A) is false as stated: at a vertex of shape $(3, 2, 1)$ one may transpose the subtrees of a single mirror pair of children,

and a parity invariant shows that this automorphism is not even a limit of compositions of the τ_T . The correct local generators are these *pair transposes*, and the corrected statement is proved equivalent to a weak form of (B) (non-mirror siblings have non-isomorphic subtrees), which holds unconditionally to depth 75. For (B), the census is equivalent to a finite vector counting the ways to peel boxes off the shape, by an identity that turns out to be Theorem 2.2 of Stanley (2003); using it we verify (B) for every shape with at most 75 boxes and prove it for rectangles, for all shapes avoiding $(3, 2, 1)$ (two-row, two-column and hook shapes), for two-corner shapes with two unit parameters, and for pairs of shapes each having at most four rows. We determine which statistics the vector provably encodes and exhibit proved families showing that neither end of the vector suffices by itself. Conjecture (B) in general, and its weak form, remain open; the shifted analogue (Problem 2) is verified to depth 60, and we prove that there is no shifted analogue of the generating-function identity with a universal factor.

Guide to the reader. The correction to Conjecture A and the corrected generators are in [Section 3](#) (Theorem 3.5 for the parity obstruction, Theorem 3.7 for the equivalence with sibling rigidity, Corollary 3.8 for the unconditional statement to depth 75). The finite reformulation of the census is Theorem 2.3 in [Section 2](#), where it is proved twice and identified with Stanley (2003), Theorem 2.2, eq. (10). The computational verification of Conjecture B to $n \leq 75$ is Theorem 4.1 in [Section 4.1](#); the classes for which B is proved are Theorems 4.4, 4.9, 4.11.2, 4.12.2 and 4.12.3; the recoverable statistics are Theorems 4.2, 4.6 and Corollary 4.10.2; the obstructions are Proposition 4.6.1, Proposition 4.10.5 and Theorem 4.13.1. The shifted tree is treated in [Section 5](#). [Section 6](#) lists open sub-questions and [Section 7](#) gives an assessment, including a list of the points a referee should check first. [Section 0](#) summarises all results with their labels.

Every statement below carries one of the labels

- **PROVED** — complete proof given here and checked line by line;
- **COMPUTATIONALLY VERIFIED** — checked by the code in `src/` over the stated range (exact integer arithmetic unless stated otherwise);
- **CONJECTURAL** — not proved.

0. Summary of results

Notation is fixed in [Section 1](#). Throughout, λ^t is the conjugate partition, f^λ the number of standard Young tableaux (SYT) of shape λ , $s_k(\lambda)$ the *up-census* (number of vertices at depth k below a vertex of shape λ in the tree SYT), and $d_j(\lambda)$ the *down-census* (number of ways to remove j boxes from λ one at a time through partitions).

1. **(PROVED; known — Stanley 2003.)** $\sum_{k \geq 0} s_k(\lambda) \frac{t^k}{k!} = e^{t+t^2/2} \sum_{j=0}^{|\lambda|} d_j(\lambda) \frac{t^j}{j!}$. Hence the in-

finite census $(s_k(\lambda))_k$ and the finite vector $(d_0(\lambda), \dots, d_{|\lambda|}(\lambda))$ determine each other, and **Conjecture B is equivalent to: the d -vector determines λ up to transpose.** The identity is Theorem 2.2, eq. (10), of R. P. Stanley, *On the enumeration of skew Young tableaux*, Adv. Appl. Math. 30 (2003) 283–294, obtained there as the exponential specialisation of the skew Littlewood identity (Macdonald, *Symmetric functions and Hall polynomials*, 2nd ed., I.5, Ex. 27(a)). We give a self-contained proof via Young’s lattice as a differential poset (the mechanism of Stanley, *Differential posets*, JAMS 1 (1988), Cor. 2.6(c)), plus the symmetric-function derivation. Verified coefficientwise for all $|\lambda| \leq 14$, $k \leq 14$ against brute-force chain enumeration.

2. **(COMPUTATIONALLY VERIFIED.)** Conjecture B holds for **all partitions of all** $n \leq 75$: for every $n \leq 75$, two partitions of n have the same d -vector only if they are equal or

conjugate. Exact integer arithmetic for $n \leq 60$ (Python), arithmetic modulo the prime $2^{61} - 1$ for $n \leq 75$ (C++; also modulo $2^{62} - 57$), which is a rigorous negative since an exact collision would persist modulo every prime. The two programs agree on all levels $n \leq 60$ (counts and an order-independent digest). Independent cross-checks of the recursion against direct skew-chain counting: all partitions of $n \leq 14$ and samples up to $n = 24$. Worley states “computation suggests” without a range; this extends it to $n \leq 75$ (8,118,264 partitions at $n = 75$).

3. **(PROVED.) Conjecture A as literally stated is false.** Let τ_T be Worley’s partial-transpose automorphism at a vertex T of symmetric shape (transpose everything in the subtree of T). The natural precise form of “every automorphism is a (possibly infinite, convergent) composition of the τ_T ” is: the closed subgroup $\overline{\langle \tau_T \rangle}$ of $\text{Aut}(\text{SYT})$ (profinite topology) is all of $\text{Aut}(\text{SYT})$. This is false: for T of shape $(3, 2, 1)$ the automorphism θ that transposes only the subtrees of the two children of shapes $(4, 2, 1)$ and $(3, 2, 1, 1)$ and fixes everything else is not in $\overline{\langle \tau_T \rangle}$. The obstruction is a family of continuous homomorphisms $E_{\lambda,p}: \text{Aut}_1 \rightarrow \mathbb{Z}/2$ (one per symmetric λ and mirror pair p of addable corners) which coincide on all τ_T but not on θ (Theorem 3.5). Confirmed by an independent group computation with sympy on the tree truncated at depth 7: the τ ’s generate a group of order 2^{27} , the pair-transposes θ generate a group of order 2^{43} , and $\theta \notin \langle \tau \rangle$.
4. **(PROVED.) Corrected Conjecture A.** Let $\theta_{T,p}$ be the *pair transpose*: for T of symmetric shape and $p = \{c, c^t\}$ an off-diagonal mirror pair of addable corners, transpose the subtrees of $T + c$ and $T + c^t$ (this swaps them) and fix everything else. Then τ_T is a product of pair transposes, and

$$\overline{\langle \theta_{T,p} \rangle} = \text{Aut}(\text{SYT}) \iff B'' \text{ (siblings with isomorphic subtrees are mirror images; Def. 3.1),}$$

and $B \Rightarrow B' \Rightarrow B''$. Under B'' every automorphism is a *unique* convergent product $\eta_0 \eta_1 \eta_2 \dots$ with η_d a product of commuting pair transposes at depth d (Theorem 3.7). **Unconditionally** (using item 2): every automorphism of SYT agrees on levels ≤ 75 with a finite product of pair transposes (Corollary 3.8).

5. **(PROVED.) Partial results on Conjecture B.** The d -vector determines: n , the number of corners r , f^λ , the quantity $m = \min_i \min(a_i, b_i)$ of the corner-run parametrisation, $A_m + B_m$ (number of corner boxes with a side of length m), the product $\prod_i P_{a_i b_i}(t)$ modulo t^{2m+1} (the “local product theorem”, Theorem 4.2), the sum of squared contents $C_2(\lambda) = \sum_{(i,j) \in \lambda} (j-i)^2$ (Theorem 4.6, via Jucys–Murphy elements), and further C_1^2 and C_4 , while the sixth entry from the top is a function of the previous ones (Section 4.10; obtained in a separate computation and independently re-verified). Conversely, pairs of shapes with equal f^λ and equal even content moments exist (first at $n = 50$), so no proof can use the top of the vector through even content moments alone. Conjecture B is further proved for two-corner shapes with at least two unit parameters (Theorem 4.11.2), and — as within-class statements — for shapes with at most three rows and, for $n \geq 21$, at most four rows (Theorems 4.12.2–4.12.3), via a new character-sum formula $n!P_\lambda(t) = \sum_{\tau \in S_n} \chi^\lambda(\tau^2)(1 + t)^{\text{fix } \tau}$ (Theorem 4.12.1). A proved infinite family of non-conjugate pairs agrees on a prefix of length $\approx n/3$ and on the top four entries simultaneously (Theorem 4.13.1), so a proof (or a counterexample) must control the middle of the vector. Consequently Conjecture B holds for all rectangles (Theorem 4.4) and — by a case analysis using explicit formulas — for all two-row shapes, two-column shapes and hooks, i.e. all shapes not containing $(3, 2, 1)$ (Theorem 4.9). A Pfaffian formula for $P_\lambda(t)$ (Theorem 4.7) gives an explicit formula for two-row shapes (Corollary 4.8). Additional results, obtained in separate computations and independently checked, are collected in Section 4.10.

6. **(CONJECTURAL.)** Conjecture B (equivalently, injectivity of $\lambda \mapsto (d_j(\lambda))_j$ up to transpose) and Conjecture B'' remain open in general. [Section 6](#) lists what would suffice.
7. Shifted analogue (Problem 2): [Section 5](#).

1. Definitions and conventions

Partitions. $\lambda = (\lambda_1 \geq \dots \geq \lambda_\ell > 0) \vdash n$, identified with its Young diagram $\{(i, j) : 1 \leq i \leq \ell, 1 \leq j \leq \lambda_i\}$ (English convention; box (i, j) = row i , column j). The *content* of box (i, j) is $c = j - i$. λ^t is the conjugate; λ is *symmetric* if $\lambda = \lambda^t$. Young's lattice $\mathbb{Y}$ is the set of partitions ordered by containment of diagrams; $\nu \leq \lambda$ means $\lambda = \nu \cup \{\text{one box}\}$. A box of λ that can be removed leaving a partition is a *removable corner*; a box outside λ that can be added is an *addable corner*. Removable corners are the boxes (i, λ_i) with $\lambda_i > \lambda_{i+1}$ (where $\lambda_{\ell+1} = 0$); addable corners are the boxes $(i, \lambda_i + 1)$ with $i = 1$ or $\lambda_{i-1} > \lambda_i$, $1 \leq i \leq \ell + 1$.

Corner-run parametrisation. Write the distinct part sizes of λ as $\alpha_1 > \dots > \alpha_r > 0$ with multiplicities $b_1, \dots, b_r \geq 1$, and put $a_i = \alpha_i - \alpha_{i+1}$ ($\alpha_{r+1} = 0$). Then $\lambda \leftrightarrow ((a_1, b_1), \dots, (a_r, b_r)) \in (\mathbb{Z}_{>0}^2)^r$ is a bijection between partitions and finite sequences of pairs of positive integers ($r = 0$ for $\emptyset$); r is the number of removable corners, the i -th corner is $c_i = (R_i, \alpha_i)$ with $R_i = b_1 + \dots + b_i$, and conjugation acts by $((a_1, b_1), \dots, (a_r, b_r)) \mapsto ((b_r, a_r), \dots, (b_1, a_1))$. The *corner region* of c_i is the $b_i \times a_i$ rectangle $\text{Reg}_i = \{(p, q) : R_{i-1} < p \leq R_i, \alpha_{i+1} < q \leq \alpha_i\} \subseteq \lambda$ ($R_0 = 0$). We write $m(\lambda) = \min_i \min(a_i, b_i)$ and $A_m = \#\{i : a_i = m\}$, $B_m = \#\{i : b_i = m\}$.

Tableaux and chains. An SYT of shape $\lambda \vdash n$ is identified with the saturated chain $\emptyset = \lambda^{(0)} < \lambda^{(1)} < \dots < \lambda^{(n)} = \lambda$ (entry i sits in $\lambda^{(i)} / \lambda^{(i-1)}$); f^λ is their number. For $\nu \subseteq \lambda$, $f^{\lambda/\nu}$ is the number of saturated chains from ν to λ (standard skew tableaux of shape λ/ν); $f^{\lambda/\nu} = 0$ if $\nu \not\subseteq \lambda$.

The tree. SYT has as vertices all SYT of all sizes (including $\emptyset$) and an edge $T - T'$ whenever T' is obtained from T by adding one box with entry $|T| + 1$. It is a tree rooted at $\emptyset$; the *depth* of T is $|T|$; the *children* of T (shape λ) are the $T + c$ for addable corners c of λ — they have pairwise distinct shapes $\lambda + c$; the *subtree* $\text{Sub}(T)$ is the set of tableaux extending T . Since $\text{Sub}(T) \cong$ the tree of saturated chains starting at λ , its rooted isomorphism type depends only on λ ; we write $\text{Sub}(\lambda)$ for this type. $\text{Aut}(\text{SYT})$ is the group of automorphisms of the unlabelled graph.

Census. For $\lambda \vdash n$ and $k, j \geq 0$:

$$s_k(\lambda) = \sum_{\mu \supseteq \lambda, |\mu|=n+k} f^{\mu/\lambda}, \quad d_j(\lambda) = \sum_{\nu \subseteq \lambda, |\lambda/\nu|=j} f^{\lambda/\nu} \quad (0 \leq j \leq n).$$

So $s_k(\lambda)$ is the number of vertices at depth $n + k$ in $\text{Sub}(T)$ for any T of shape λ , and $d_j(\lambda)$ the number of ways to remove j boxes one at a time through partitions. Clearly $d_0 = 1$, $d_1 = r$, $d_j(\lambda) = \sum_c d_{j-1}(\lambda - c)$ (sum over removable corners), $d_n = d_{n-1} = f^\lambda$, and (since every chain from (1) to λ passes through (2) or $(1, 1)$) $d_{n-2} = f^{\lambda/(2)} + f^{\lambda/(1,1)} = f^{\lambda/(1)} = f^\lambda$. We put

$$E_\lambda(t) = \sum_{k \geq 0} s_k(\lambda) \frac{t^k}{k!}, \quad P_\lambda(t) = \sum_{j=0}^n d_j(\lambda) \frac{t^j}{j!}, \quad u_i(\lambda) = d_{n-i}(\lambda) = \sum_{\nu \vdash i} f^{\lambda/\nu}.$$

Both census vectors are invariant under $\lambda \mapsto \lambda^t$ (transposition is an automorphism of $\mathbb{Y}$).

Filters. A subset $S \subseteq \lambda$ is a *filter* (order filter of the poset λ with $(i, j) \leq (i', j')$ iff $i \leq i'$, $j \leq j'$) if $u \in S$, $v \in \lambda$, $u \leq v$ imply $v \in S$; equivalently $\lambda \setminus S$ is a partition $\nu \subseteq \lambda$. Removing

the boxes of $S = \lambda/\nu$ one at a time through partitions means removing at each step a maximal element of what is left of S ; hence

$$d_j(\lambda) = \sum_{S \text{ filter of } \lambda, |S|=j} e(S), \quad e(S) = \# \text{linear extensions of the poset } S. \quad (1.1)$$

Involution numbers. $I_k = \#\{\text{SYT with } k \text{ boxes}\} = \#\{\text{involutions of } [k]\}$, $\sum_k I_k t^k / k! = e^{t+t^2/2}$, $I_k = I_{k-1} + (k-1)I_{k-2}$.

The conjectures (our formulation of the two halves of Worley's Problem 3):

- **A.** Every automorphism of SYT is a composition of the partial transposes τ_T (Definition 3.2). Precise form fixed in [Section 3.3](#).
- **B.** $(s_k(\lambda))_{k \geq 0} = (s_k(\mu))_{k \geq 0} \Rightarrow \mu \in \{\lambda, \lambda^t\}$.
- **B'.** $\text{Sub}(\lambda) \cong \text{Sub}(\mu)$ (rooted trees) $\Rightarrow \mu \in \{\lambda, \lambda^t\}$.
- **B''.** For every λ and distinct addable corners $c \neq c'$ of λ : $\text{Sub}(\lambda + c) \cong \text{Sub}(\lambda + c') \Rightarrow \lambda = \lambda^t$ and $c' = c^t$ (Definition 3.1).

$B \Rightarrow B' \Rightarrow B''$: the census is an invariant of the rooted isomorphism type, and by Lemma 3.4 below $\lambda + c' \in \{\lambda + c, (\lambda + c)^t\}$ with $c \neq c'$ forces $\lambda = \lambda^t$, $c' = c^t$.

2. The generating-function identity

2.1 Two facts about Young's lattice

Lemma 2.1 (PROVED). For every partition λ , $\#\{\text{addable corners}\} = \#\{\text{removable corners}\} + 1$.

Proof. In the corner-run parametrisation the removable corners are $c_1, \dots, c_r$. The addable corners are $(1, \alpha_1 + 1)$, the boxes $(R_{i-1} + 1, \alpha_i + 1)$ for $2 \leq i \leq r$ (the first row of the i -th run, which is shorter than the row above it), and $(R_r + 1, 1)$: in total $1 + (r-1) + 1 = r + 1$. For $\lambda = \emptyset$: 0 and 1. $\square$

Lemma 2.2 (PROVED). Let $\lambda \neq \mu$ be partitions of the same size n . Then $\#\{\nu \vdash n+1 : \nu \succ \lambda, \nu \succ \mu\} = \#\{\nu \vdash n-1 : \nu \preceq \lambda, \nu \preceq \mu\} \in \{0, 1\}$.

Proof. As sets of boxes, $\nu \supseteq \lambda \cup \mu$ with $|\nu| = n+1$ forces $\nu = \lambda \cup \mu$ and $|\lambda \cup \mu| = n+1$; conversely if $|\lambda \cup \mu| = n+1$ then $\lambda \cup \mu$, a union of two order ideals of $\mathbb{N}^2$, is a partition covering both. Likewise a common lower cover exists iff $|\lambda \cap \mu| = n-1$, and then it is $\lambda \cap \mu$. Finally $|\lambda \cup \mu| + |\lambda \cap \mu| = 2n$. $\square$

Let V be the free abelian group on $\mathbb{Y}$, $U, D: V \rightarrow V$ the up and down operators $U\lambda = \sum_{\mu \succ \lambda} \mu$, $D\lambda = \sum_{\nu \preceq \lambda} \nu$, and $\omega: V \rightarrow \mathbb{Z}$ the linear functional with $\omega(\lambda) = 1$ for all λ . Lemmas 2.1-2.2 say exactly (comparing coefficients of μ in $DU\lambda$ and $UD\lambda$):

$$DU - UD = I \quad \text{on } V, \quad \omega \circ U = \omega \circ D + \omega. \quad (2.1)$$

(That is, $\mathbb{Y}$ is a 1-differential poset in the sense of Stanley.) By definition of chains,

$$s_k(\lambda) = \omega(U^k \lambda), \quad d_j(\lambda) = \omega(D^j \lambda). \quad (2.2)$$

2.2 The identity

Theorem 2.3 (PROVED; Stanley 2003, Thm 2.2 (10)). For every partition λ ,

$$E_\lambda(t) = e^{t+t^2/2} P_\lambda(t), \quad \text{i.e.} \quad s_k(\lambda) = \sum_{j=0}^{\min(k, |\lambda|)} \binom{k}{j} I_{k-j} d_j(\lambda).$$

Proof. From $DU = UD + I$ one gets by induction $DU^k = U^k D + kU^{k-1}$ for $k \geq 1$ (indeed $DU^k = (UD + I)U^{k-1} = U(U^{k-1}D + (k-1)U^{k-2}) + U^{k-1}$). Apply $\omega \circ U = \omega \circ D + \omega$ to the vector $U^k \lambda$:

$$\omega(U^{k+1} \lambda) = \omega(DU^k \lambda) + \omega(U^k \lambda) = \omega(U^k D \lambda) + k \omega(U^{k-1} \lambda) + \omega(U^k \lambda).$$

Writing $A_v(t) = \sum_k \omega(U^k v) t^k / k!$ for $v \in V$ (linear in v) and multiplying by $t^k / k!$, summing over $k \geq 0$:

$$A'_\lambda(t) = A_{D\lambda}(t) + t A_\lambda(t) + A_\lambda(t), \quad A_{D\lambda} = \sum_{\nu < \lambda} A_\nu.$$

Put $B_\lambda(t) = e^{-t-t^2/2} A_\lambda(t)$. Then $B'_\lambda = e^{-t-t^2/2} (A'_\lambda - (1+t)A_\lambda) = e^{-t-t^2/2} A_{D\lambda} = \sum_{\nu < \lambda} B_\nu$, and $B_\lambda(0) = A_\lambda(0) = 1$. The polynomials P_λ satisfy the same system: $P'_\lambda = \sum_{\nu < \lambda} P_\nu$ (from $d_j(\lambda) = \sum_c d_{j-1}(\lambda - c)$) and $P_\lambda(0) = 1$. By induction on $|\lambda|$: for $\lambda = \emptyset$, $D\emptyset = 0$ so $B'_\emptyset = P'_\emptyset = 0$ and $B_\emptyset = P_\emptyset = 1$; for general λ , $B'_\lambda = \sum_\nu B_\nu = \sum_\nu P_\nu = P'_\lambda$ by induction and $B_\lambda(0) = P_\lambda(0)$, so $B_\lambda = P_\lambda$ as formal power series. Hence $E_\lambda = A_\lambda = e^{t+t^2/2} P_\lambda$. The coefficient form follows by multiplying the series. $\square$

Remarks. (i) The same proof shows that in an r -differential poset, $\sum_k \omega(U^k x) t^k / k! = e^{r(t+t^2/2)} \sum_j \omega(D^j x) t^j / j!$ for every element x ; this is the fixed-start-vertex form of Stanley (1988), Cor. 2.6(c). (ii) Sanity checks: $\lambda = \emptyset$ gives $E_\emptyset = e^{t+t^2/2}$ (the EGF of I_k), and $\lambda = (1)$ gives $(1+t)e^{t+t^2/2} = \frac{d}{dt} e^{t+t^2/2}$.

Second derivation (symmetric functions; sketch, after Stanley 2003). With the Hall inner product $\langle p_\rho, p_\sigma \rangle = z_\rho \delta_{\rho\sigma}$ one has $\langle p_1 f, g \rangle = \langle f, \partial g / \partial p_1 \rangle$ (check on the p -basis: $z_{\rho \cup 1} = z_\rho (m_1(\rho) + 1)$), hence $\partial s_\lambda / \partial p_1 = \sum_{\nu < \lambda} s_\nu$ (adjoint of the Pieri rule $p_1 s_\nu = \sum_{\mu > \nu} s_\mu$). Let $F = \sum_\mu s_\mu = \prod_i (1 - x_i)^{-1} \prod_{i < j} (1 - x_i x_j)^{-1} = \exp(\sum_{k \geq 1} \frac{p_k}{k} + \sum_{k \geq 1} \frac{p_k^2 - p_{2k}}{2k})$ (Littlewood). Then $s_k(\lambda) = \langle s_\lambda p_1^k, F \rangle$, so $E_\lambda(t) = \langle s_\lambda, F(p_1 + t) \rangle$; the only p_1 -dependence of $\log F$ is $p_1 + p_1^2/2$, so $F(p_1 + t) = F \cdot e^{t+t^2/2} e^{tp_1}$, and $\langle s_\lambda, p_1^j F \rangle = \langle \partial^j s_\lambda / \partial p_1^j, F \rangle = \sum_{|\lambda/\nu|=j} f^{\lambda/\nu} \langle s_\nu, F \rangle = d_j(\lambda)$. This also shows

$$d_j(\lambda) = \langle s_\lambda, p_1^j F \rangle, \quad \sum_\lambda P_\lambda(t) s_\lambda = e^{tp_1} F. \quad (2.3)$$

Computational verification. `src/verify_identity.py`: $s_k(\lambda)$ computed by brute-force enumeration of upward chains (`census.s_up`, memoised recursion on the definition) and independently by enumerating all $\mu \vdash n+k$ and counting skew chains (`s_up_by_skew`), compared with $\sum_j \binom{k}{j} I_{k-j} d_j(\lambda)$: **identity holds for all** $|\lambda| \leq 14, k \leq 14$ (`data/identity_check_n14_k14.txt`), and the two routes for s_k agree for $|\lambda| \leq 6, k \leq 5$.

Corollary 2.4 (PROVED). $(s_k(\lambda))_{k \geq 0} = (s_k(\mu))_{k \geq 0}$ iff $(d_j(\lambda))_j = (d_j(\mu))_j$ (as finite sequences; in particular $|\lambda| = \deg P_\lambda = |\mu|$). Hence **Conjecture B** $\iff$ **the d -vector determines λ up to transpose.** $\square$

3. Automorphisms of the tree

3.1 Basic facts

Lemma 3.0 (PROVED). Every graph automorphism of SYT fixes $\emptyset$, preserves depth, and maps children of T to children of $\varphi(T)$; φ restricted to $\text{Sub}(T)$ is an isomorphism of rooted trees onto $\text{Sub}(\varphi(T))$.

Proof. $\emptyset$ has degree 1 (its only neighbour is (1)); a nonempty vertex of shape λ has one parent and $r + 1 \geq 2$ children (Lemma 2.1), so degree ≥ 3 . Thus $\emptyset$ is the unique vertex of degree 1 and is fixed; depth is the distance to $\emptyset$; the parent of $T \neq \emptyset$ is its unique neighbour of smaller depth. $\square$

Definition 3.2 (partial transposes). Let T be a vertex of symmetric shape λ , $n = |T|$. Every $V \in \text{Sub}(T)$ is $V = T \cup S$ with S a standard skew tableau of shape μ/λ (entries $n + 1, \dots, |V|$), and S^t is a standard skew tableau of shape $\mu^t/\lambda^t = \mu^t/\lambda$. Put

$$\tau_T(V) = T \cup S^t \quad (V \in \text{Sub}(T)), \quad \tau_T(V) = V \text{ otherwise.}$$

For an off-diagonal mirror pair $p = \{c, c^t\}$ of addable corners of λ (i.e. $c = (i, j)$ addable with $i \neq j$; then $c^t = (j, i)$ is addable too, by symmetry) put

$$\theta_{T,p}(V) = \tau_T(V) \text{ if } V \in \text{Sub}(T + c) \cup \text{Sub}(T + c^t), \quad \theta_{T,p}(V) = V \text{ otherwise.}$$

Lemma 3.3 (PROVED; the warm-up). τ_T and $\theta_{T,p}$ are involutive automorphisms of SYT. Moreover $\tau_T = \prod_p \theta_{T,p} \cdot \prod_{c=c^t} \tau_{T+c}$ (products over the off-diagonal mirror pairs and over the at most one diagonal addable corner; all factors commute).

Proof. τ_T is a bijection of $\text{Sub}(T)$ ($S \mapsto S^t$ is an involution) fixing T and all vertices outside $\text{Sub}(T)$. If $V = T \cup S \in \text{Sub}(T)$, $V \neq T$, its parent is $T \cup S'$ where S' is S minus its largest entry, and τ_T maps the edge $V - \text{parent}(V)$ to $T \cup S^t - T \cup (S')^t$, which is an edge since $(S')^t = (S^t)'$; the edge $T - \text{parent}(T)$ and all edges outside $\text{Sub}(T)$ are fixed. So τ_T preserves the edge set. For $\theta_{T,p}$: τ_T maps $\text{Sub}(T + c)$ onto $\text{Sub}(T + c^t)$ (the entry $n + 1$ of S sits at c iff that of S^t sits at c^t), so $\theta_{T,p}$ is a bijection; edges inside the two subtrees are preserved as before, the edges $T - (T + c)$ and $T - (T + c^t)$ are swapped, all other edges are fixed. For the factorisation: $\text{Sub}(T) \setminus \{T\}$ is the disjoint union of the $\text{Sub}(T + c)$ over addable corners c ; on $\text{Sub}(T + c) \cup \text{Sub}(T + c^t)$ ($c \neq c^t$) τ_T agrees with $\theta_{T,p}$; on $\text{Sub}(T + c)$ with $c = c^t$ (necessarily $c = (k + 1, k + 1)$ where k is the Durfee size, so at most one such c) the shape $\lambda + c$ is symmetric and $\tau_T(T + c \cup S') = T + c \cup (S')^t = \tau_{T+c}(T + c \cup S')$. $\square$

Lemma 3.4 (siblings; PROVED). Let $c \neq c'$ be addable corners of λ with $(\lambda + c)^t = \lambda + c'$. Then $\lambda = \lambda^t$ and $c' = c^t$.

Proof. $\lambda \subseteq \lambda + c' = (\lambda + c)^t$, so (transposing) $\lambda^t \subseteq \lambda + c$; also $\lambda^t \subseteq \lambda^t + c^t = (\lambda + c)^t = \lambda + c'$. If $\lambda \neq \lambda^t$ then $\lambda \cup \lambda^t$ has more than $|\lambda|$ boxes and is contained in both $\lambda + c$ and $\lambda + c'$, which have $|\lambda| + 1$ boxes; so $\lambda + c = \lambda \cup \lambda^t = \lambda + c'$, contradicting $c \neq c'$. Hence $\lambda = \lambda^t$, and then $\lambda + c' = (\lambda + c)^t = \lambda^t + c^t = \lambda + c^t$ gives $c' = c^t$. $\square$

3.2 The parity obstruction: the literal Conjecture A is false

Topology. $\text{Aut}(\text{SYT})$ is the inverse limit of its images in $\text{Aut}(\text{SYT}_{\leq N})$ ($N \geq 0$), where $\text{SYT}_{\leq N}$ is the truncation to depth $\leq N$; give it the profinite topology (pointwise convergence). A subset is closed iff it is determined by finitely many levels at a time; “ $g_N \rightarrow g$ ” iff for every N , g_N eventually agrees with g on levels $\leq N$. The natural precise version of “convergent composition

of τ 's" is membership in the closure $\overline{\langle \tau_T \rangle}$ of the subgroup generated by all τ_T ; equivalently: for every N the restriction to levels $\leq N$ agrees with a finite product of τ 's. (Any weaker reading — e.g. arbitrary infinite products — is contained in this one, because any limit of finite products lies in the closure.)

Let $\text{Aut}_1 \subseteq \text{Aut}(\text{SYT})$ be the set of automorphisms g with $\text{shape}(g(V)) \in \{\text{shape}(V), \text{shape}(V)^t\}$ for all vertices V . It is a closed subgroup (the condition is a conjunction of conditions each involving one level) containing every τ_T and every $\theta_{T,p}$.

Definition (parity homomorphisms). Fix a symmetric $\lambda \vdash d$ and an off-diagonal mirror pair $p = \{c, c^t\}$ of addable corners of λ . For $g \in \text{Aut}_1$ and T of shape λ : $g(T)$ has shape λ (as $\lambda^t = \lambda$), and $g(T + c)$ is a child of $g(T)$ of shape $\lambda + c$ or $(\lambda + c)^t = \lambda + c^t$, hence $g(T + c) \in \{g(T) + c, g(T) + c^t\}$. Put $\sigma_g(T, p) = 1$ if $g(T + c) = g(T) + c^t$ and 0 otherwise (this is symmetric in $c \leftrightarrow c^t$ because g is injective on the children of T), and

$$E_{\lambda,p}(g) = \sum_{T \text{ of shape } \lambda} \sigma_g(T, p) \pmod{2}.$$

Theorem 3.5 (PROVED). 1. $E_{\lambda,p} : \text{Aut}_1 \rightarrow \mathbb{Z}/2$ is a continuous group homomorphism. 2. For T_0 of symmetric shape λ_0 : $E_{\lambda,p}(\tau_{T_0}) = f^{\lambda/\lambda_0} \pmod{2}$ if $\lambda_0 \subseteq \lambda$, and 0 otherwise — **independent of p** . For T of shape λ and a pair p' : $E_{\lambda,p}(\theta_{T,p'}) = [p = p']$. 3. Consequently, if λ is symmetric with two distinct off-diagonal mirror pairs $p_1 \neq p_2$ (the smallest example is $\lambda = (3, 2, 1)$ with pairs $\{(1, 4), (4, 1)\}, \{(2, 3), (3, 2)\}$), then for any T of shape λ , $\theta_{T,p_1} \in \text{Aut}(\text{SYT}) \setminus \overline{\langle \tau_{T'} : T' \rangle}$.

Conjecture A in its literal form is false, and $\overline{\langle \tau \rangle} \subsetneq \overline{\langle \theta \rangle} \subseteq \text{Aut}(\text{SYT})$.

Proof. (1) For $g, h \in \text{Aut}_1$: $h(T + c) = h(T) + c_1$ with $c_1 = c$ or c^t according to $\sigma_h(T, p)$, and $g(h(T) + c_1) = gh(T) + c_2$ with $c_2 = c_1$ or c_1^t according to $\sigma_g(h(T), p)$; so $\sigma_{gh}(T, p) = \sigma_h(T, p) + \sigma_g(h(T), p) \pmod{2}$. Summing over all T of shape λ and using that $T \mapsto h(T)$ permutes the SYT of shape λ gives $E_{\lambda,p}(gh) = E_{\lambda,p}(h) + E_{\lambda,p}(g)$. Continuity: $E_{\lambda,p}(g)$ depends only on the action of g on levels $d, d + 1$. (2) τ_{T_0} fixes every $T \not\supseteq T_0$ together with its children, so $\sigma = 0$ there. If $T = T_0 \cup S \supseteq T_0$ then $\tau_{T_0}(T) = T_0 \cup S^t$ and $\tau_{T_0}(T + c) = T_0 \cup (S + c)^t = (T_0 \cup S^t) + c^t$, so $\sigma_{\tau_{T_0}}(T, p) = 1$ for every pair p . The number of SYT T of shape λ containing the fixed tableau T_0 is f^{λ/λ_0} . For $\theta_{T,p'}$: it moves only vertices strictly below T , so the only tableau of shape λ with $\sigma \neq 0$ is T itself, where $\sigma(T, p) = [p = p']$. (3) By (2), E_{λ,p_1} and E_{λ,p_2} agree on every generator $\tau_{T'}$, hence (being homomorphisms) on $\overline{\langle \tau \rangle}$, hence (being continuous, and $\{g : E_{\lambda,p_1}(g) = E_{\lambda,p_2}(g)\}$ being closed in Aut_1) on $\overline{\langle \tau \rangle} \subseteq \text{Aut}_1$. But $E_{\lambda,p_1}(\theta_{T,p_1}) = 1 \neq 0 = E_{\lambda,p_2}(\theta_{T,p_1})$. Finally $\tau_T \in \overline{\langle \theta \rangle}$ restricted to any finite depth by Lemma 3.3 and induction on depth, so $\overline{\langle \tau \rangle} \subseteq \overline{\langle \theta \rangle}$. $\square$

Computational confirmation (src/automorphism_check.py, src/tree.py; sympy PermutationGroup). On $\text{SYT}_{\leq 7}$ (352 vertices): all 28 τ_T ($|\overline{T}| \leq 6$, symmetric shape) and all 43 $\theta_{T,p}$ are edge-preserving; $E_{(3,2,1),p_1}(\tau_T) = E_{(3,2,1),p_2}(\tau_T)$ for all 28 τ_T ; $E_{(3,2,1),p_1}(\theta_{T,p_1}) = 1$, $E_{(3,2,1),p_2}(\theta_{T,p_1}) = 0$; $|\langle \tau \rangle_7| = 2^{27}$, $|\langle \theta \rangle_7| = 2^{43}$; every $\tau_T \in \langle \theta \rangle_7$; and Schreier-Sims membership testing returns $\theta_{T,p_1} \notin \langle \tau \rangle_7$. (At depths 5 and 6 the two groups coincide, orders 2^5 and 2^{11} ; the first symmetric shape with two off-diagonal mirror pairs is $(3, 2, 1)$ at depth 6. Note also $\tau_\emptyset = \tau_{(1)}$, which is why 28 generators give order 2^{27} .)

Remark. This is the same phenomenon as in the Young-Fibonacci tree (Zenodo record 21538865 on Worley's Problem 9, where it is remarked that "not every automorphism is a

finite product of local swaps"); here the extra twist is that even the *closure* of the group generated by Worley's whole-subtree transposes is too small, and the correct local generators are the pair transposes.

3.3 The corrected Conjecture A and its equivalence with B''

Definition 3.1 (B''). For every partition λ and distinct addable corners $c \neq c'$: $\text{Sub}(\lambda + c) \cong \text{Sub}(\lambda + c')$ implies $\lambda = \lambda^t$ and $c' = c^t$.

Conjecture A _{θ} . $\text{Aut}(\text{SYT}) = \overline{\langle \theta_{T,p} \rangle}$, i.e. for every automorphism φ and every N there is a finite product of pair transposes agreeing with φ on levels $\leq N$.

Theorem 3.7 (PROVED). $A_\theta \iff B''$. Moreover, if B'' holds then every $\varphi \in \text{Aut}(\text{SYT})$ can be written uniquely as a convergent product $\varphi = \lim_N \eta_0 \eta_1 \cdots \eta_N$ where $\eta_d = \prod_{(T,p) \in \Pi_d} \theta_{T,p}$ for a set Π_d of pairs (vertex T at depth d of symmetric shape, off-diagonal mirror pair p) — the factors of each η_d commute — and every choice of sets $(\Pi_d)_{d \geq 0}$ arises. In particular $\text{Aut}(\text{SYT})$ is then an inverse limit of finite 2-groups, of cardinality $2^{\aleph_0}$, and $\text{Aut}(\text{SYT}) = \text{Aut}_1$.

Proof. ($B'' \Rightarrow A_\theta$ and the product decomposition.) Let $\varphi \in \text{Aut}(\text{SYT})$. We construct $\psi_d \in \langle \theta \rangle$ with $\psi_d = \varphi$ on levels $\leq d$, $\psi_0 = \text{id}$, $\psi_{d+1} = \psi_d \eta_d$. Suppose ψ_d is constructed and let $\chi = \psi_d^{-1} \varphi$; χ is an automorphism that is the identity on levels $\leq d$. Let T be a vertex at depth d of shape λ and c an addable corner. Since χ fixes T it permutes the children of T , and χ maps $\text{Sub}(T + c)$ isomorphically onto $\text{Sub}(\chi(T + c))$ (Lemma 3.0). If $\chi(T + c) = T + c'$ with $c' \neq c$ then $\text{Sub}(\lambda + c) \cong \text{Sub}(\lambda + c')$, so by B'' , $\lambda = \lambda^t$ and $c' = c^t$. Hence: if λ is not symmetric, χ fixes all children of T ; if λ is symmetric, χ fixes the child at the diagonal corner (if any) and, for each off-diagonal mirror pair $p = \{c, c^t\}$, either fixes $T + c$ and $T + c^t$ or swaps them. Let Π_d be the set of (T, p) (T at depth d , symmetric shape) for which χ swaps the pair p , and $\eta_d = \prod_{(T,p) \in \Pi_d} \theta_{T,p}$; the factors have pairwise disjoint supports (different T 's at the same depth have disjoint subtrees; different pairs at the same T involve disjoint pairs of subtrees), so they commute and the product is well defined. By construction η_d is the identity on levels $\leq d$ and agrees with χ on level $d + 1$. Then $\psi_{d+1} := \psi_d \eta_d$ agrees with φ on levels $\leq d + 1$: for V of depth $\leq d + 1$, $\psi_d \eta_d(V) = \psi_d \chi(V) = \varphi(V)$. Thus $\psi_N \rightarrow \varphi$, so $\varphi \in \overline{\langle \theta \rangle}$, and $\varphi = \lim \eta_0 \cdots \eta_N$. *Uniqueness and freeness.* Given any sets (Π_d) , the partial products $\eta_0 \cdots \eta_N$ stabilise on each level (factors of depth $\geq N$ act trivially on levels $\leq N$), so they converge to an automorphism φ ; and Π_d is recovered from φ as the set of pairs swapped by $(\eta_0 \cdots \eta_{d-1})^{-1} \varphi$ at depth d — note that this uses only φ on levels $\leq d + 1$ and $\Pi_0, \dots, \Pi_{d-1}$. So $(\Pi_d)_d \mapsto \varphi$ is a bijection from $\prod_d 2^{\mathcal{P}_d}$ ($\mathcal{P}_d$ = the set of pairs at depth d) onto $\text{Aut}(\text{SYT})$; since $\mathcal{P}_d \neq \emptyset$ for infinitely many d (e.g. every staircase $(k, k - 1, \dots, 1)$, $k \geq 2$, is symmetric with the off-diagonal pair $\{(1, k + 1), (k + 1, 1)\}$), the cardinality is $2^{\aleph_0}$. Every pair transpose lies in Aut_1 and Aut_1 is closed, so $\text{Aut}(\text{SYT}) = \overline{\langle \theta \rangle} \subseteq \text{Aut}_1$. ($A_\theta \Rightarrow B''$.) Suppose B'' fails: $\lambda, c \neq c'$ with $\text{Sub}(\lambda + c) \cong \text{Sub}(\lambda + c')$ but not ($\lambda = \lambda^t$ and $c' = c^t$). Take any T of shape λ and a rooted-tree isomorphism $\iota: \text{Sub}(T + c) \rightarrow \text{Sub}(T + c')$; define $g = \iota$ on $\text{Sub}(T + c)$, $g = \iota^{-1}$ on $\text{Sub}(T + c')$, $g = \text{id}$ elsewhere. Then g is an automorphism (it preserves edges inside the two subtrees, swaps the edges $T - (T + c)$, $T - (T + c')$, and fixes all others), and $g(T + c) = T + c'$ has shape $\lambda + c' \notin \{\lambda + c, (\lambda + c)^t\}$ by Lemma 3.4. So $g \notin \text{Aut}_1 \supseteq \overline{\langle \theta \rangle}$, i.e. A_θ fails. $\square$

Corollary 3.8 (PROVED, with computer verification). Let $N \leq 75$. Every automorphism of SYT agrees on levels $\leq N$ with a finite product of pair transposes $\theta_{T,p}$ with $|T| < N$. Equivalently, the image of $\text{Aut}(\text{SYT})$ in $\text{Aut}(\text{SYT}_{\leq N})$ equals the image of $\langle \theta \rangle$, a 2-group of order $2^{\#\{\text{pairs at depth} < N\}}$.

Proof. The inductive step in the proof of Theorem 3.7 at depth d uses B'' only for shapes $\lambda \vdash d$,

i.e. only the statement “distinct non-mirror siblings $\lambda + c, \lambda + c' \vdash d + 1$ have non-isomorphic subtrees”. By [Section 4.1](#) (Conjecture B verified for all $n \leq 75$) two partitions of $d + 1 \leq 75$ with isomorphic subtrees have equal census, hence are equal or conjugate, hence by Lemma 3.4 are mirror siblings. So the construction of $\psi_1, \dots, \psi_N$ goes through for $N \leq 75$. The order statement follows from the uniqueness part (the sets $\Pi_0, \dots, \Pi_{N-1}$ are free parameters and determine the action on levels $\leq N$). $\square$

Consequences for the shape of the answer to Problem 3. Assuming only B”, $\text{Aut}(\text{SYT})$ is exactly the group of “independent local swaps”: at each vertex of symmetric shape one may swap, independently for each off-diagonal mirror pair of children, the two subtrees by transposition. Worley’s τ_T are the products over all pairs at T (and over the diagonal child); they generate a proper closed subgroup, of infinite index (the homomorphisms $E_{\lambda, p_1} - E_{\lambda, p_2}$ for the infinitely many symmetric λ with ≥ 2 off-diagonal pairs are independent and vanish on it).

4. Conjecture B

4.1 Computations

`src/dcensus_scan.py` (Python, exact) computes level by level $(d_j(\lambda))_j$ for all $\lambda \vdash n$ from level $n - 1$ via $d_j(\lambda) = \sum_c d_{j-1}(\lambda - c)$, groups the partitions of n by d -vector and reports every group that is not $\{\lambda\}$ (λ symmetric) or $\{\lambda, \lambda^t\}$. `src/dcensus_modp.cpp` does the same modulo a prime ($2^{61} - 1$ or $2^{62} - 57$), ranking partitions in reverse-lexicographic order.

Theorem 4.1 (COMPUTATIONALLY VERIFIED). For every $n \leq 75$ and all $\lambda, \mu \vdash n$: $d(\lambda) = d(\mu) \Rightarrow \mu \in \{\lambda, \lambda^t\}$. Hence Conjecture B, B’ and B” hold for all shapes of size ≤ 75 (B” for $|\lambda| \leq 74$).

Evidence. (i) Exact, $n \leq 60$: `data/collisions_python.txt` (per n : $p(n)$, number of distinct d -vectors, number of symmetric shapes, number of transpose pairs, number of collision groups = 0, SHA-256 of the sorted list of d -vectors, and the digest $\sum_{\lambda, j} d_j(\lambda) 7^j \bmod (2^{61} - 1)$); at every n the number of distinct vectors equals $(\# \text{symmetric}) + (\# \text{transpose pairs})$. (ii) Modulo $2^{61} - 1$, $n \leq 75$: `data/scan_modp_p0_n75.txt`, zero collisions; the counts and digests agree with (i) for every $n \leq 60$. Modulo $2^{62} - 57$: `data/scan_modp_p1.txt` ($n \leq 60$) and `data/scan_modp_p1_n75.txt`. Since $d(\lambda) = d(\mu)$ over $\mathbb{Z}$ implies congruence modulo every prime, zero collisions modulo one prime already proves the statement. (iii) Independent checks of the recursion: `src/crosscheck_skew.py` recomputes d_j from the definition (enumerating all $\nu \subseteq \lambda$ and counting skew chains) for all partitions of $n \leq 14$ and 25 random partitions of $n \in [20, 24]$: 533 partitions, no mismatch; `tests/test_basic.py` checks hand-computed values. (iv) `src/independent_check.py` is a from-scratch reimplementing sharing no code with the library: partitions in multiplicity representation with corners read off the distinct part sizes reproduce the counts, collision counts and mod- $(2^{61} - 1)$ digests of the main scan for every $n \leq 40$ (`data/independent_check_n40.txt`); and for $n \leq 24$ the d -vectors recovered from up-chain counts through the identity $P_\lambda = e^{-t-t^2/2} E_\lambda$ (a different recursion, using addable rather than removable corners) agree with the down recursion for every partition. Full d -vectors for $n \leq 30$ are stored in `data/dvectors_nXX.txt.gz`. Running time: 6 min (Python, $n \leq 60$) and a few minutes (C++, $n \leq 75$, ≈ 9 GB RAM).

4.2 The local product theorem

For a rectangle a^b (i.e. b rows of length a) put $G_{a,b}(t) = P_{a^b}(t)$.

Lemma 4.1 (PROVED). $d_j(a^b) = \sum_{\rho \vdash j, \ell(\rho) \leq b, \rho_1 \leq a} f^\rho$ for all j ; i.e. $G_{a,b}(t) = \sum_{\rho \subseteq b \times a} f^\rho t^{|\rho|} / |\rho|!$.

In particular $G_{a,b} = G_{b,a}$ and $G_{a,b} \equiv e^{t+t^2/2} \pmod{t^{\min(a,b)+1}}$.

Proof. A filter S of the rectangle a^b is the complement of a partition $\nu \subseteq a^b$; rotating by 180° about the centre of the rectangle, S becomes the diagram of a partition ρ with $\ell(\rho) \leq b$, $\rho_1 \leq a$, and the rotation reverses the order, so $e(S) = e(\rho) = f^\rho$. Apply (1.1). The last claim: every $\rho \vdash j \leq \min(a, b)$ fits in the box, and $\sum_{\rho \vdash j} f^\rho = I_j$. $\square$

Theorem 4.2 (local product theorem; PROVED). Let λ have corner runs $((a_i, b_i))_{i=1}^r$ and $m = \min_i \min(a_i, b_i)$. Then

$$P_\lambda(t) \equiv \prod_{i=1}^r G_{a_i, b_i}(t) \pmod{t^{2m+1}},$$

i.e. $d_j(\lambda) = j! [t^j] \prod_i G_{a_i, b_i}(t)$ for $0 \leq j \leq 2m$. In particular 1. $d_j(\lambda) = j! [t^j] e^{r(t+t^2/2)}$ for $0 \leq j \leq m$; 2. $d_{m+1}(\lambda) = (m+1)! [t^{m+1}] e^{r(t+t^2/2)} - (A_m + B_m)$, and $A_m + B_m \geq 1$; 3. the d -vector determines n, r, m (as one less than the first index j with $d_j \neq j! [t^j] e^{r(t+t^2/2)}$; such $j \leq n$ exists for $n \geq 2$), $A_m + B_m$, and the polynomial $\prod_i G_{a_i, b_i} \pmod{t^{2m+1}}$, which depends only on the multiset of unordered pairs $\{a_i, b_i\}$; 4. $d_2(\lambda) = r(r-1) + \#\{i : a_i \geq 2\} + \#\{i : b_i \geq 2\}$.

Proof. Let S be a filter of λ with $|S| \leq 2m$. We use the following facts about the poset S (induced from $\mathbb{N}^2$). (a) *The order on S is generated by adjacency inside S .* If $u = (i, j) \leq v = (i', j')$ are in S , then the boxes $(i, j), (i, j+1), \dots, (i, j')$ lie in λ (row i has length $\geq j'$ since $(i', j') \in \lambda$, $i \leq i'$) and hence in S (filter), and likewise $(i, j'), \dots, (i', j')$; this is a path of adjacent boxes in S from u to v . Consequently the connected components of S (adjacency) are pairwise incomparable, and S is the disjoint union, as a poset, of its components $K_1, \dots, K_s$; hence $e(S) = \prod_{i=1}^s e(K_i)$. (b) *Every maximal element of S is a removable corner of λ* (if $(i, j+1) \in \lambda$ it lies in S and is larger; likewise $(i+1, j)$). Every component K is a finite nonempty poset, so it contains a maximal element of S , i.e. a corner c_i of λ . (c) *A component contains only one corner.* A path of adjacent boxes from $c_i = (R_i, \alpha_i)$ to $c_{i'}$, $i < i'$, has at least $(R_{i'} - R_i) + (\alpha_i - \alpha_{i'}) + 1 \geq b_{i+1} + a_i + 1 \geq 2m + 1 > |S|$ boxes. (d) *The component K of c_i lies in Reg_i and is a filter of the rectangle Reg_i with unique maximal element c_i .* By (b), (c), c_i is the unique maximal element of K , so $K \subseteq \{(p, q) : p \leq R_i, q \leq \alpha_i\}$. If $(p, q) \in K$ with $p \leq R_{i-1}$ ($i \geq 2$), then, S being a filter, the adjacent boxes $(p, q), (p, q+1), \dots, (p, \lambda_p)$ all lie in S , so $(p, \lambda_p) \in K$ with $\lambda_p \geq \alpha_{i-1} > \alpha_i$, contradicting maximality of c_i ; symmetrically $q > \alpha_{i+1}$. So $K \subseteq \text{Reg}_i$. Moving right from a box of Reg_i within λ stays in Reg_i (rows $R_{i-1} < p \leq R_i$ have length α_i), and moving down from row R_i leaves λ (row $R_i + 1$ has length $\alpha_{i+1} < q$); hence K , being closed under moving right/down in λ , is a filter of the rectangle Reg_i , i.e. by the proof of Lemma 4.1 the 180° -rotation of a partition $\rho_i \subseteq b_i \times a_i$, with $e(K) = f^{\rho_i}$. (e) *Conversely*, for any $(\rho_1, \dots, \rho_r)$ with $\rho_i \subseteq b_i \times a_i$, the union S of the rotated ρ_i 's placed at the corners is a filter of λ (each piece is closed under right/down moves in λ by the argument in (d)), and its components are exactly the nonempty pieces (boxes of Reg_i and $\text{Reg}_{i'}$ are never adjacent: consecutive regions occupy disjoint column ranges and disjoint row ranges). By (a)-(e), for $j \leq 2m$,

$$d_j(\lambda) = \sum_{\substack{(\rho_i)_i, \rho_i \subseteq b_i \times a_i \\ \sum |\rho_i| = j}} \binom{j}{|\rho_1|, \dots, |\rho_r|} \prod_i f^{\rho_i} = j! [t^j] \prod_i \left(\sum_{\rho \subseteq b_i \times a_i} f^\rho \frac{t^{|\rho|}}{|\rho|!} \right) = j! [t^j] \prod_i G_{a_i, b_i}(t).$$

(1) For $j \leq m$ every $\rho \vdash j$ fits in every $b_i \times a_i$ box, so each factor is $e^{t+t^2/2}$ modulo t^{m+1} . (2) $G_{a_i, b_i} \equiv e^{t+t^2/2} - ([a_i = m] + [b_i = m]) \frac{t^{m+1}}{(m+1)!} \pmod{t^{m+2}}$, because the only partitions of

$m + 1$ not fitting in a box with both sides $\geq m$ are $(m + 1)$ (iff the width is m) and (1^{m+1}) (iff the height is m), each with $f = 1$; multiply out modulo t^{m+2} . $A_m + B_m \geq 1$ since some corner attains the minimum. (3) $n = \deg P_\lambda$, $r = d_1$; by (1),(2) d_j agrees with $j![t^j]e^{r(t+t^2/2)}$ exactly for $j \leq m$ and fails at $j = m + 1$ (which is $\leq n$ for $n \geq 2$: $m^2 \leq a_1 b_1 \leq n$ and $m + 1 \leq \sqrt{n} + 1 \leq n$ for $n \geq 3$; $n = 2$: $m = 1$). $A_m + B_m$ is the deficit at $m + 1$, and $G_{a,b} = G_{b,a}$. (4) is the case $m = 1$, $j = 2$ of the formula: $2![t^2]e^{r(t+t^2/2)} = r^2 + r$, minus $A_1 + B_1$. $\square$

Computational verification (src/explore_separation.py): the congruence $P_\lambda \equiv \prod G_{a_i, b_i} \pmod{t^{2m+1}}$, statement (1) and the deficit formula (2) were checked for all $|\lambda| \leq 18$.

Remark. The bound $2m$ is sharp in general: for $j = 2m + 1$ a single component can contain two adjacent corners (the “rim path” of length $a_i + b_{i+1} + 1$ when $a_i = b_{i+1} = m$). Beyond $j = 2m$ the coefficients depend on the *order* of the corners and on the orientation (a_i, b_i) vs (b_i, a_i) , which is exactly what distinguishes non-conjugate shapes with the same multiset of corner boxes.

4.3 Rectangles

Theorem 4.4 (PROVED). Conjecture B holds for rectangles: if $d(\mu) = d(a^b)$ then $\mu \in \{a^b, b^a\}$.

Proof. $d_1(\mu) = 1$, so μ is a rectangle $a'^{b'}$ with $a'b' = ab = n$. By Theorem 4.2(3), $m = \min(a, b) = \min(a', b')$ and $A_m + B_m = 1 + [a = b] = 1 + [a' = b']$. If $a = b$ then $a' = b' = m$. Otherwise $\{a, b\} = \{m, n/m\} = \{a', b'\}$. $\square$

4.4 Statistics at the top of the vector

Recall $u_i = d_{n-i} = \sum_{\nu \vdash i} f^{\lambda/\nu}$ and $u_0 = u_1 = u_2 = f^\lambda$.

Lemma 4.5 (PROVED). For $\nu \vdash k \leq n$, $f^{\lambda/\nu} = \langle s_\lambda, s_\nu p_1^{n-k} \rangle$, and for a partition $\rho \vdash k$, $\langle s_\lambda, p_\rho p_1^{n-k} \rangle = \chi^\lambda(\rho \cup 1^{n-k})$ (irreducible character of S_n at the cycle type $\rho \cup 1^{n-k}$). Consequently

$$u_i = \sum_{\rho \vdash i} \frac{\sigma(\rho)}{z_\rho} \chi^\lambda(\rho \cup 1^{n-i}), \quad \sigma(\rho) = \sum_{\nu \vdash i} \chi^\nu(\rho) = \#\{\tau \in S_i : \tau^2 = g\} \text{ for any fixed } g \text{ of cycle type } \rho,$$

and $u_3 = \frac{2}{3}f^\lambda + \frac{1}{3}\chi^\lambda(3, 1^{n-3})$, $u_4 = \frac{5}{12}f^\lambda + \frac{1}{4}\chi^\lambda(2, 2, 1^{n-4}) + \frac{1}{3}\chi^\lambda(3, 1^{n-3})$, $u_5 = \frac{13}{60}f^\lambda + \frac{1}{4}\chi^\lambda(2, 2, 1^{n-4}) + \frac{1}{3}\chi^\lambda(3, 1^{n-3}) + \frac{1}{5}\chi^\lambda(5, 1^{n-5})$.

Proof. $s_\nu p_1^{n-k} = \sum_\mu f^{\mu/\nu} s_\mu$ by the Pieri rule iterated; $p_\rho = \sum_\nu \chi^\nu(\rho) s_\nu$ (Frobenius); $s_\nu = \sum_\rho z_\rho^{-1} \chi^\nu(\rho) p_\rho$. The number of square roots formula $\sum_\nu \chi^\nu(\pi) = \#\{\tau : \tau^2 = \pi\}$ is the classical Frobenius-Schur count (all χ^ν are real with indicator 1). The explicit cases: $\sigma(1^3) = 4$, $\sigma(3) = 1$, $\sigma(2, 1) = 0$; $\sigma(1^4) = 10$, $\sigma(2, 2) = 2$, $\sigma(3, 1) = 1$, $\sigma(4) = \sigma(2, 1, 1) = 0$; $\sigma(1^5) = 26$, $\sigma(2, 2, 1) = 2$, $\sigma(3, 1, 1) = 2$, $\sigma(5) = 1$, others 0 (a permutation is a square iff its even cycle lengths occur an even number of times). $\square$

Theorem 4.6 (PROVED). Let $n \geq 3$ and $C_2(\lambda) = \sum_{(i,j) \in \lambda} (j - i)^2$. Then

$$3d_{n-3}(\lambda) - 2f^\lambda = \chi^\lambda(3, 1^{n-3}) = f^\lambda \cdot \frac{3C_2(\lambda) - \frac{3}{2}n(n-1)}{n(n-1)(n-2)}.$$

Hence the d -vector determines $C_2(\lambda)$.

Proof. The first equality is Lemma 4.5 ($u_3 = \frac{2}{3}f + \frac{1}{3}\chi(3\text{-cycle})$; equivalently, from $f = f^{\lambda/3} + 2f^{\lambda/21} + f^{\lambda/111}$, $u_3 = f^{\lambda/3} + f^{\lambda/21} + f^{\lambda/111}$ and $p_3 = s_3 - s_{21} + s_{111}$). For the second, use the Jucys-Murphy elements $J_k = \sum_{i < k} (i k) \in \mathbb{Q}S_n$ ($k = 1, \dots, n$). Classical theorem (Jucys, Murphy; see Okounkov-Vershik): in the irreducible representation V_λ with its Young (seminormal) basis $\{v_T\}$ indexed by SYT T of shape λ , $J_k v_T = c_T(k) v_T$ where $c_T(k)$ is the content of the box of T containing k . Hence $\text{tr}_{V_\lambda} \sum_k J_k^2 = \sum_T \sum_k c_T(k)^2 = f^\lambda C_2(\lambda)$. On the other hand, in the group algebra, $J_k^2 = (k-1)e + \sum_{i \neq i' < k} (i k)(i' k)$ and $(i k)(i' k)$ is a 3-cycle on $\{i, i', k\}$; the ordered pairs (i, i') give each of the two 3-cycles on a 3-set with maximum k exactly once. Summing over k : $\sum_k J_k^2 = \binom{n}{2}e + Z_3$ with Z_3 the sum of all 3-cycles. Taking traces on V_λ : $f^\lambda C_2 = \binom{n}{2}f^\lambda + 2\binom{n}{3}\chi^\lambda(3, 1^{n-3})$, which rearranges to the formula. $\square$

Computational verification (src/check_formulas.py): both equalities checked for all $3 \leq |\lambda| \leq 16$ (with $\chi^\lambda(3\text{-cycle})$ computed independently as $f^{\lambda/3} - f^{\lambda/21} + f^{\lambda/111}$).

Remarks. (i) The same method gives, from u_4 and u_5 , the character values at $(2, 2, 1^{n-4})$ and $(5, 1^{n-5})$, which are polynomials in n , C_1^2 , C_2 , C_4 ($C_k = \sum c^k$); values at odd permutations (e.g. transpositions, C_1 itself) are *not* determined, consistent with $C_k(\lambda^t) = (-1)^k C_k(\lambda)$. From u_7 on, each u_i mixes at least two new even classes, so this route yields one new content polynomial per degree only up to degree 6. (ii) Since the content multiset determines λ , and the multiset $\{|c|\}$ is transpose-invariant, one may ask whether the *even* content power sums $C_2, C_4, \dots$ determine λ up to transpose; the d -vector is not known to determine them all. (iii) src/explore_invariants.py: the pair (f^λ, C_2) alone does **not** separate non-conjugate partitions (first collisions at $n = 14$); nor does (r, f^λ) or $(d_1, d_2, d_3, f^\lambda)$.

4.5 A Pfaffian formula

Theorem 4.7 (PROVED). Let $\ell \geq \ell(\lambda)$ be even, pad λ with zeros to length ℓ , and set $\alpha_i = \lambda_i + \ell - i$. Then

$$P_\lambda(t) = \text{Pf}(B_{ij}(t))_{1 \leq i, j \leq \ell}, \quad B_{ij}(t) = [x^{\alpha_i} y^{\alpha_j}] \frac{(x-y)e^{t(x+y)}}{(1-xy)(1-x)(1-y)},$$

and $[x^p y^q] \frac{x-y}{(1-xy)(1-x)(1-y)} = \text{sgn}(p-q)$, so $B_{ij}(t) = \sum_{a \leq \alpha_i, b \leq \alpha_j} \text{sgn}(\alpha_i - a - \alpha_j + b) \frac{t^{a+b}}{a! b!}$.

Proof. By (2.3), $P_\lambda(t)$ is the coefficient of s_λ in $e^{tp_1} F$. For any symmetric function g and $\ell \geq \ell(\lambda)$, the coefficient of s_λ in g equals $[x^{\lambda+\delta}] a_\delta(x_1, \dots, x_\ell) g(x_1, \dots, x_\ell)$ with $\delta = (\ell-1, \dots, 0)$ and $a_\delta = \prod_{i < j} (x_i - x_j)$ (because $a_\delta s_\mu = a_{\mu+\delta}$, $[x^{\lambda+\delta}] a_{\mu+\delta} = \delta_{\lambda\mu}$, and $s_\mu(x_1, \dots, x_\ell) = 0$ for $\ell(\mu) > \ell$). In ℓ variables $e^{tp_1} F = \prod_i \frac{e^{tx_i}}{1-x_i} \prod_{i < j} \frac{1}{1-x_i x_j}$, so

$$P_\lambda(t) = [x^\alpha] \prod_{i < j} \frac{x_i - x_j}{1 - x_i x_j} \prod_i \phi(x_i), \quad \phi(x) = \frac{e^{tx}}{1-x}.$$

Schur's Pfaffian identity, in the form $\text{Pf}(\frac{x_i - x_j}{1 - x_i x_j})_{i, j \leq \ell} = \prod_{i < j} \frac{x_i - x_j}{1 - x_i x_j}$ for even ℓ , follows from the classical $\text{Pf}(\frac{y_i - y_j}{y_i + y_j}) = \prod_{i < j} \frac{y_i - y_j}{y_i + y_j}$ by the substitution $x_i = \frac{1 - y_i}{1 + y_i}$, under which $\frac{x_i - x_j}{1 - x_i x_j} = -\frac{y_i - y_j}{y_i + y_j}$ and the signs $(-1)^{\ell/2}$, $(-1)^{\ell(\ell-1)/2}$ agree for even ℓ . Since $\text{Pf}(\Delta M \Delta) = \det(\Delta) \text{Pf}(M)$ for diagonal Δ , the product equals $\text{Pf}(\frac{x_i - x_j}{1 - x_i x_j} \phi(x_i) \phi(x_j))$; expanding the Pfaffian as a signed sum

over perfect matchings, each term is a product of factors involving disjoint pairs of variables, so coefficient extraction $[x^\alpha]$ factors over the pairs, giving $\text{Pf}(B)$. Finally $\frac{1}{(1-xy)(1-x)(1-y)} = \sum_{p,q} (\min(p,q) + 1)x^p y^q$, and multiplying by $x - y$ gives the coefficient $(\min(p-1, q) + 1)[p \geq 1] - (\min(p, q-1) + 1)[q \geq 1] = \text{sgn}(p - q)$ (check the cases $p > q$, $p = q$, $p < q$). $\square$

Computational verification (src/check_formulas.py, sympy): the Pfaffian formula was checked symbolically in t for all partitions with ≤ 2 rows and $|\lambda| \leq 7$ ($\ell = 2$) and all with ≤ 4 rows and $|\lambda| \leq 7$ ($\ell = 4$). Schur's identity in the form $\text{Pf}((x_i - x_j)/(1 - x_i x_j)) = \prod_{i < j} (x_i - x_j)/(1 - x_i x_j)$ was checked symbolically for $\ell = 2, 4, 6$ (src/check_schur_pfaffian.py).

Corollary 4.8 (two-row shapes; PROVED). For $\lambda = (\lambda_1, \lambda_2)$, $\delta = \lambda_1 - \lambda_2 \geq 0$, and $0 \leq j \leq n$:

$$d_j(\lambda_1, \lambda_2) = \sum_{b=0}^{\min(j, \lambda_2)} \binom{j}{b} \text{sgn}(\delta + 1 - j + 2b) [j - b \leq \lambda_1 + 1].$$

In particular $d_j = 2^j$ for $j \leq \min(\delta, \lambda_2)$, and $d_j \leq 2^j$ always.

Proof. Take $\ell = 2$: $P_\lambda = B_{12}$ with $\alpha = (\lambda_1 + 1, \lambda_2)$, and read off $j![t^j]$ with $a = j - b$. $\square$

Remark. Combinatorially, a removal sequence from a two-row shape is a word in $\{\text{row } 1, \text{row } 2\}$ and the formula is a reflection-principle count; the Pfaffian formula produces it mechanically. For hooks $(a, 1^b)$ one has directly $d_j = \sum_i \binom{j}{i} [i \leq a - 1][j - i \leq b]$ for $j < n$ and $d_n = \binom{n-1}{b}$ (the removal sequence is a word in $\{\text{arm}, \text{leg}\}$ and the box $(1, 1)$ is removed last).

4.6 The hard pairs

src/explore_separation.py computes, for each $n \leq 22$, the largest index j_0 at which two non-conjugate partitions of n first differ. For $7 \leq n \leq 22$ the maximum is $j_0 = n - 3$, attained by $(2, 2, 1^{n-4})$ vs $(3, 1^{n-3})$. This is an instance of a family:

Proposition 4.6.1 (PROVED). Let $m \geq 2$ and $\lambda_1 > m$, $n = \lambda_1 + m$. The two-row shape (λ_1, m) and the hook $(\lambda_1, 1^m)$ satisfy $d_j(\lambda_1, m) = d_j(\lambda_1, 1^m)$ for all $0 \leq j \leq n - 2m$, and $d_{n-2m+1}(\lambda_1, 1^m) - d_{n-2m+1}(\lambda_1, m) = 1$.

Proof. For $j \leq \delta = \lambda_1 - m$, Corollary 4.8 gives $d_j(\lambda_1, m) = \sum_{b \leq \min(j, m)} \binom{j}{b}$ (all signs are + since $\delta + 1 - j + 2b \geq 1$, and $j - b \leq \lambda_1 + 1$), and (4.9.1) gives $d_j(\lambda_1, 1^m) = \sum_i \binom{j}{i} [i \leq \lambda_1 - 1][j - i \leq m] = \sum_{b \leq \min(j, m)} \binom{j}{b}$ because $j \leq \lambda_1 - 1$. At $j = \delta + 1$ the two-row sum loses its $b = 0$ term (sign 0) while the hook sum keeps it (as $j \leq \lambda_1 - 1$ since $m \geq 2$). $\square$

Separating depth (COMPUTATIONALLY VERIFIED). src/separating_depth.py: for every $6 \leq n \leq 45$, the prefix $(d_0, \dots, d_{n-3})$ already separates all transpose classes of partitions of n , while $(d_0, \dots, d_{n-4})$ does not (Proposition 4.6.1). By Theorem 2.3 this says that the up-census must be known to depth exactly $n - 3$ below a vertex of size n to identify its shape — and that depth suffices for $n \leq 45$. So there is no bounded-depth invariant, but the *last* informative coordinate d_{n-3} (equivalently f^λ and $C_2(\lambda)$, Theorem 4.6) always finishes the job in the verified range.

For $m = 2$ this is $(n - 2, 2)$ vs $(3, 1^{n-3})$, i.e. (transposing the first) $(2, 2, 1^{n-4})$ vs $(3, 1^{n-3})$: **identical d_j for all $j \leq n - 4$** , different only in $d_{n-3}, \dots, d_n$ ($f^{(3, 1^{n-3})} = \binom{n-1}{2}$, $f^{(2, 2, 1^{n-4})} = \frac{n(n-3)}{2}$). So a proof of Conjecture B cannot rely on any bounded initial segment of the vector, nor (by Section 4.2) on the local corner data alone; the top entries $u_0 = f^\lambda$, u_3 (i.e. C_2) are essential. Conversely (Remark 4.6(iii)) the top entries alone are not enough either, and Theorem 4.13.1 gives an infinite family agreeing at both ends at once.

4.7 Reformulation via shifted Schur functions (for orientation)

By the Okounkov-Olshanski theory, $(n)_k f^{\lambda/\nu}/f^\lambda = s_\nu^*(\lambda)$ ($\nu \vdash k$, $(n)_k = n(n-1)\cdots(n-k+1)$) is the value at λ of the shifted Schur function s_ν^* . Hence

$$\frac{n!}{f^\lambda} P_\lambda(t) = \sum_{i=0}^n g_i(\lambda) t^{n-i}, \quad g_i = \sum_{\nu \vdash i} s_\nu^* \in \Lambda^*,$$

and Conjecture B says that f^λ together with the values $g_1(\lambda), \dots, g_n(\lambda)$ of these transpose-invariant shifted symmetric functions determine λ up to transpose. Since Λ^* is generated by the content power sums C_k and transposition acts by $C_k \mapsto (-1)^k C_k$, the transpose-invariant subalgebra needs more than one generator per degree from degree 8 on, whereas the g_i provide one function per degree; so a proof cannot proceed by showing that the g_i generate the invariant subalgebra — it must use the integrality/discreteness of λ (or the full d -vector in a different way). (This paragraph is orientation only; nothing below depends on it.)

4.8 What is recoverable — summary

From $(d_j(\lambda))_{j=0}^n$ one recovers (PROVED): n ; r (Thm 4.2); f^λ ; m , $A_m + B_m$, and $\prod_i G_{a_i, b_i} \bmod t^{2m+1}$ (Thm 4.2); $C_2(\lambda)$ (Thm 4.6), C_1^2 and C_4 (Cor. 4.10.2), χ^λ at $(2, 2, 1^{n-4})$, $(5, 1^{n-5})$, $(3, 3, 1^{n-6})$ (Lemma 4.5), and d_{n-6} is determined by the other top entries (Thm 4.10.3). Conjecture B is PROVED for rectangles (Thm 4.4), for two-row/two-column/hook shapes (Thm 4.9) and for the classes in [Section 4.10](#), and COMPUTATIONALLY VERIFIED for all $n \leq 75$.

4.9 Shapes with no three-corner subshape: two-row, two-column and hook shapes

Let $\mathcal{C} = \{\lambda : (3, 2, 1) \not\subseteq \lambda\} = \{\lambda : \ell(\lambda) \leq 2\} \cup \{\lambda : \lambda_1 \leq 2\} \cup \{\lambda : \lambda_2 \leq 1\}$ — the two-row shapes, the two-column shapes and the hooks. (A shape with three removable corners has three distinct part sizes $\alpha_1 > \alpha_2 > \alpha_3 \geq 1$, hence rows of lengths $\geq 3, \geq 2, \geq 1$, hence contains $(3, 2, 1)$; so every subshape of a member of $\mathcal{C}$ has at most two corners, and conversely.)

Theorem 4.9 (PROVED). Conjecture B holds for every $\lambda \in \mathcal{C}$: if $\mu \vdash n = |\lambda|$ and $d(\mu) = d(\lambda)$, then $\mu \in \{\lambda, \lambda^t\}$.

Throughout, “the formulas” refers to Corollary 4.8 for two-row shapes and to

$$d_j(a, 1^b) = \sum_{i=0}^j \binom{j}{i} [i \leq a-1] [j-i \leq b] \quad (j < n), \quad d_n(a, 1^b) = \binom{n-1}{b} \quad (n = a+b) \quad (4.9.1)$$

for hooks (a removal sequence from $(a, 1^b)$ that does not remove the box $(1, 1)$ is an arbitrary word in {arm, leg} with at most $a-1$ arm letters and at most b leg letters; the box $(1, 1)$ can only be removed last, and $d_n = f^{(a, 1^b)} = \binom{n-1}{b}$ by the hook length formula). Both formulas were checked against the recursion for all such shapes with $n \leq 40$ (src/explore_classC.py); every numbered identity below was checked for all parameters with $n \leq 34$ (src/check_classC_cases.py).

Proof. Since $d(\lambda) = d(\lambda^t)$ and the conclusion is symmetric, we may replace λ by λ^t and μ by μ^t at will.

Step 1 (one corner). $r(\mu) = d_1(\mu) = d_1(\lambda) = r(\lambda) \leq 2$. If $r = 1$ both are rectangles and Theorem 4.4 applies. So let $r = 2$; then λ is a two-row shape (λ_1, λ_2) with $\lambda_1 > \lambda_2 \geq 1$, or a two-column shape, or a hook $(a, 1^b)$ with $a \geq 2, b \geq 1$.

Step 2 (d_2 forces μ into six families). By Theorem 4.2(4), for $r = 2$, $d_2 = 6 - A_1 - B_1$ with $A_1 = \#\{i : a_i = 1\}$, $B_1 = \#\{i : b_i = 1\}$. A two-row λ has runs $((\delta, 1), (\lambda_2, 1))$ ($\delta = \lambda_1 - \lambda_2$),

a hook $(a, 1^b)$ has runs $((a-1, 1), (1, b))$, so $A_1 + B_1 \geq 2$ and $d_2(\lambda) \leq 4$ for $\lambda \in \mathcal{C}$. Hence $A_1(\mu) + B_1(\mu) \geq 2$, and writing the runs of μ as $((a_1, b_1), (a_2, b_2))$, one of the six coincidences $b_1 = b_2 = 1, a_1 = a_2 = 1, a_2 = b_1 = 1, a_1 = b_1 = 1, a_1 = b_2 = 1, a_2 = b_2 = 1$ holds, i.e. μ is respectively a two-row shape, a two-column shape, a hook $(a_1 + 1, 1^{b_2})$, or one of

$$F_1(c, d) = (c+1, c^d), \quad F_2(b, c) = ((c+1)^b, c), \quad F_4(a, b) = ((a+1)^b, 1) = F_1(b, a)^t.$$

$F_1(c, d)$ with $d = 1$ or $c = 1$, $F_2(b, c)$ with $b = 1$ or $c = 1$, and $F_4(a, b)$ with $b = 1$ or $a = 1$ lie in $\mathcal{C}$; so either $\mu \in \mathcal{C}$ or μ is $F_1(c, d)$, $F_2(b, c)$ or $F_4(a, b)$ with both parameters ≥ 2 .

Step 3 (d_3 excludes the F -families). For $\lambda \in \mathcal{C}$, $d_3(\lambda) \leq 8$ (both formulas are sums of at most 2^3 terms of absolute value $\binom{3}{i}$ with signs, and in fact $d_j \leq 2^j$). Compute $d_3(\mu) = \sum_c d_2(\mu - c)$ for the F -shapes with parameters ≥ 2 , using $d_2(\nu) = r(r-1) + \#\{i : a_i \geq 2\} + \#\{i : b_i \geq 2\}$ (Theorem 4.2(4), valid for all ν): $*F_1(c, d) - c_1 = (c^{d+1})$, runs $((c, d+1))$: $d_2 = 2$; $F_1(c, d) - c_2 = (c+1, c^{d-1}, c-1)$, runs $((1, 1), (1, d-1), (c-1, 1))$, $r = 3$: $d_2 = 6 + [c \geq 3] + [d \geq 3]$. So $d_3(F_1(c, d)) = 8 + [c \geq 3] + [d \geq 3]$. $*F_2(b, c) - c_1 = ((c+1)^{b-1}, c, c)$, runs $((1, b-1), (c, 2))$: $d_2 = 4 + [b \geq 3]$; $F_2(b, c) - c_2 = ((c+1)^b, c-1)$, runs $((2, b), (c-1, 1))$: $d_2 = 4 + [c \geq 3]$. So $d_3(F_2(b, c)) = 8 + [b \geq 3] + [c \geq 3]$. $*d_3(F_4(a, b)) = d_3(F_1(b, a)) = 8 + [a \geq 3] + [b \geq 3]$.

Thus $d_3(\mu) \geq 9 > d_3(\lambda)$ unless $\mu \in \{F_1(2, 2), F_2(2, 2), F_4(2, 2)\} = \{(3, 2, 2), (3, 3, 2), (3, 3, 1)\}$, with $n \in \{7, 8\}$ and $d_3(\mu) = 8$. In that case $d_3(\lambda) = 8$; from the formulas, a two-row shape has $d_3 = 8$ only if $\delta \geq 3$ and $\lambda_2 \geq 3$ (so $n \geq 9$), and a hook $(a, 1^b)$ only if $a \geq 4, b \geq 3$, so λ or λ^t is $(4, 1^3)$ ($n = 7$) or one of $(5, 1^3), (4, 1^4)$ ($n = 8$), with $f^\lambda \in \{20, 35\}$, whereas $f^{(3, 2, 2)} = f^{(3, 3, 1)} = 21$ and $f^{(3, 3, 2)} = 42$ (hook length formula) — contradicting $d_n(\mu) = d_n(\lambda)$. Hence $\mu \in \mathcal{C}$.

Step 4. Now $\lambda, \mu \in \mathcal{C}$ with $r = 2$; transposing, we may assume each of λ, μ is a two-row shape with $\lambda_2 \geq 1$ or a hook $(a, 1^b)$ with $a \geq 2, b \geq 1$. For a two-row shape put $m_\lambda = \min(\delta, \lambda_2)$; for a hook put $m_\mu = \min(a-1, b)$. From the formulas:

$$d_j = 2^j \ (j \leq m), \quad d_{m+1} = 2^{m+1} - 1 - \varepsilon, \quad \varepsilon = [\delta = \lambda_2] \text{ resp. } [a-1 = b], \quad (4.9.2)$$

(for the two-row shape at $j \leq m$ all $b \leq j \leq \lambda_2$ occur with positive sign since $\delta + 1 - j + 2b > 0$; at $j = m+1$ exactly the term $b = m+1$ is missing if $\lambda_2 = m$, and exactly the term $b = 0$ has sign 0 if $\delta = m$; for the hook at $j \leq m$ no constraint is active, and at $j = m+1$ exactly the term $i = m+1$ is lost if $a-1 = m$ and exactly $i = 0$ if $b = m$; in both cases $m+1 < n$). So $m := m_\lambda = m_\mu$ and $\varepsilon_\lambda = \varepsilon_\mu$ are determined by the d -vector. Next, the value d_{m+2} ($m+2 \leq n$ in all cases below):

$$\begin{aligned} \text{T1: } \lambda_2 = m < \delta : & \quad d_{m+2} = 2^{m+2} - m - 3 - [\delta = m+1], \\ \text{T2: } \delta = m < \lambda_2 : & \quad d_{m+2} = 2^{m+2} - 2 - [\lambda_2 = m+1], \\ \text{T3: } \delta = \lambda_2 = m : & \quad d_{m+2} = 2^{m+2} - m - 5, \\ \text{H1: } a-1 = m < b : & \quad d_{m+2} = 2^{m+2} - m - 3 - [b = m+1], \\ \text{H2: } b = m < a-1 : & \quad d_{m+2} = 2^{m+2} - m - 3 - [a-1 = m+1], \\ \text{H3: } a-1 = b = m : & \quad d_{m+2} = 2^{m+2} - 2m - 6. \end{aligned} \quad (4.9.3)$$

(T1: $b \leq m$, and $\text{sgn}(\delta - m - 1 + 2b) = +1$ except 0 when $b = 0, \delta = m+1$; the missing terms are $b = m+1, m+2$. T2: all $b \leq \min(m+2, \lambda_2)$ occur, sign -1 for $b = 0$ and $+1$ otherwise, and $b = m+2$ is missing iff $\lambda_2 = m+1$. T3: $b \leq m$, signs as in T2. H1: lost $i = m+1, m+2$, and $i = 0$ iff $b = m+1$. H2: lost $i = 0, 1$, and $i = m+2$ iff $a-1 = m+1$. H3: lost $i = 0, 1, m+1, m+2$.)

(4a) *Two two-row shapes.* ε tells whether $\delta = \lambda_2$. If so, $\lambda = (2m, m)$ is determined by m (and $n = 3m$). Otherwise λ is $(n-m, m)$ (case T1) or $(\frac{n+m}{2}, \frac{n-m}{2})$ (case T2), and by (4.9.3) these

two have different d_{m+2} : equality would need $m+1+[n=3m+1]=[n=3m+2]$, impossible for $m \geq 1$. So $\mu = \lambda$.

(4b) *Two hooks.* $f^\lambda = \binom{n-1}{b} = \binom{n-1}{b'} = f^\mu$, and $k \mapsto \binom{n-1}{k}$ is strictly increasing on $0 \leq k \leq \frac{n-1}{2}$ and symmetric about $\frac{n-1}{2}$; so $b' \in \{b, n-1-b\}$, i.e. $\mu \in \{\lambda, \lambda^t\}$ (the transpose of $(a, 1^b)$ is $(b+1, 1^{a-1})$, with $\text{leg } n-1-b$).

(4c) *A two-row shape $\lambda = (\lambda_1, \lambda_2)$ and a hook $\mu = (a, 1^b)$.* If $\lambda_2 = 1$ then λ is a hook and (4b) applies; if $b = 1$ then μ is two-row and (4a) applies. So let $\lambda_2 \geq 2$, $b \geq 2$; we derive a contradiction from $d(\lambda) = d(\mu)$. By (4.9.2), $m_\lambda = m_\mu = m$ and $[\delta = \lambda_2] = [a-1=b]$, so the case of λ in $\{T1, T2, T3\}$ and of μ in $\{H1, H2, H3\}$ match as $T3 \leftrightarrow H3$ and $\{T1, T2\} \leftrightarrow \{H1, H2\}$. By (4.9.3): T3 vs H3 would need $m+5 = 2m+6$; T2 vs H2 would need $m+1+[a-1=m+1] = [\lambda_2 = m+1]$; T2 vs H1 would need $m+1+[b=m+1] = [\lambda_2 = m+1]$ — all impossible for $m \geq 1$. There remain T1 vs H1 and T1 vs H2, where $\lambda = (\lambda_1, m)$ with $m = \lambda_2 \geq 2$. In case H2, $\mu = (a, 1^m)$ with $a = \lambda_1$ (same n); in case H1, $\mu = (m+1, 1^b)$ with $b = \lambda_1 - 1$, and $\mu^t = (\lambda_1, 1^m)$. Either way $d(\lambda) = d((\lambda_1, 1^m))$, and we compare at $j = \delta+1 = \lambda_1 - m + 1$ (note $2 \leq m$ gives $j \leq \lambda_1 - 1 < n$): by Corollary 4.8, $d_j(\lambda_1, m) = \sum_{1 \leq b' \leq m} \binom{j}{b'}$ (the term $b' = 0$ has sign $\text{sgn}(\delta+1-j) = 0$, all others sign $+1$, and $j-b' \leq \lambda_1+1$ holds), while by (4.9.1), with $i \leq \lambda_1 - 1$ automatic and $j-i \leq m$, $d_j(\lambda_1, 1^m) = \sum_{i \geq j-m} \binom{j}{i} = \sum_{0 \leq b' \leq m} \binom{j}{b'}$. The two differ by $\binom{j}{0} = 1$. Contradiction. $\square$

Remarks. (i) The proof only uses d_1, d_2, d_3 , the entries d_j for $j \leq m+2$ and $j = \delta+1$, and f^λ ; it is a case analysis rather than a structural argument, and does not obviously extend to general two-corner shapes $((a+c)^b, c^d)$, for which no closed form of the d -vector is available. (ii) An independent proof of the same theorem was obtained separately and checked by two independent reviewers (scripts `src/agents/classC_verify.py`, `src/agents/classC_extra.py`); it proves Corollary 4.8 directly by a reflection-principle argument on words in $\{\text{row } 1, \text{row } 2\}$ and settles the last case by $f^{(\lambda_1, m)} = \binom{n-1}{m} - \binom{n-1}{m-2} < \binom{n-1}{m} = f^{(\lambda_1, 1^m)}$ instead of the entry $d_{\delta+1}$. The two proofs were obtained independently and agree in every formula.

4.10 What the top of the vector determines (re-verified)

A separate computation (scripts `src/agents/topend_*.py`) extended Theorem 4.6 systematically. Its proofs are elementary — explicit class expansions in $\mathbb{Q}[S_n]$ of power sums of the Jucys–Murphy elements, plus the fact that these act by contents — and everything below that is marked PROVED has a complete proof that was checked line by line; the formulas were independently re-verified with separately written code in `src/check_topend.py` (Murnaghan–Nakayama characters for Theorem 4.10.1, the d -vector recursion for Theorems 4.10.2–4.10.3, hook lengths and contents for the separation data). Notation: for a partition ρ without parts 1, K_ρ is the class sum of cycle type $\rho \cup 1^{n-|\rho|}$ and $\omega_\lambda(K_\rho) = |K_\rho| \chi^\lambda(\rho \cup 1^{n-|\rho|}) / f^\lambda$ is its central character, $|K_\rho| = (n)_{|\rho|} / z_\rho$; $C_k = \sum_{\square} c(\square)^k$.

Theorem 4.10.1 (PROVED; formulas re-verified for all $\lambda \vdash n \leq 12$). With $J_k = \sum_{i < k} (i \ k)$ one has in $\mathbb{Q}[S_n]$: $\sum_k J_k = K_2$, $\sum_k J_k^2 = \binom{n}{2} + K_3$, $K_2^2 = \binom{n}{2} + 3K_3 + 2K_{22}$, $\sum_k J_k^4 = K_5 + (3n-4)K_3 + 4K_{22} + \frac{n(n-1)(4n-5)}{6}$, $K_3^2 = 2\binom{n}{3} + (3n-8)K_3 + 8K_{22} + 5K_5 + 2K_{33}$ (the last three by explicit enumeration of coincidence patterns; all five checked in $\mathbb{Z}[S_n]$ for $n \leq 8$). Since symmetric polynomials in the J_k act on V_λ by the same polynomials in the contents,

$$\begin{aligned} \omega_\lambda(K_3) &= C_2 - \binom{n}{2}, & \omega_\lambda(K_{22}) &= \frac{1}{2}C_1^2 - \frac{3}{2}C_2 + \binom{n}{2}, \\ \omega_\lambda(K_5) &= C_4 - (3n-10)C_2 - 2C_1^2 + \frac{n(n-1)(5n-19)}{6}, \end{aligned}$$

and $\omega_\lambda(K_{33})$ is an explicit polynomial in n, C_1^2, C_2, C_4 .

Corollary 4.10.2 (PROVED). The d -vector determines C_2, C_1^2 and C_4 : by Lemma 4.5, $\chi^\lambda(2, 2, 1^{n-4}) = 4u_4 - 4u_3 + f$ and $\chi^\lambda(5, 1^{n-5}) = 5u_5 - 5u_4 + f$, and Theorem 4.10.1 converts these (triangularly) into C_2 , then C_1^2 , then C_4 . (Lemma 4.5 also gives $\chi^\lambda(3, 3, 1^{n-6}) = \frac{9}{2}u_6 - \frac{9}{2}u_5 + \frac{3}{2}u_3 - \frac{1}{2}f$; the coefficient table of u_6 is $\frac{19}{180}f + \frac{2}{9}\chi(3) + \frac{1}{4}\chi(2, 2) + \frac{1}{5}\chi(5) + \frac{2}{9}\chi(3, 3)$.)

Theorem 4.10.3 (level 6 is redundant; PROVED; re-verified). For $n \geq 6$,

$$\frac{(n)_6 d_{n-6}(\lambda)}{f^\lambda} = (n^2 - 11n + 42)C_1^2 + 2C_2^2 + \left(\frac{2n^3}{3} - 16n^2 + \frac{328n}{3} - 210\right)C_2 \\ + (n - 15)C_4 + \frac{n(n-1)(19n^4 - 326n^3 + 2489n^2 - 9856n + 15390)}{180},$$

so d_{n-6} is a function of $(n, d_n, d_{n-3}, d_{n-4}, d_{n-5})$. (The polynomial as originally computed omitted the pure- n term; it was recovered by interpolation, and the corrected identity was verified on all partitions of $6 \leq n \leq 22$, the functional dependence on all partitions of $n \leq 30$.) *Proof.* Insert the u_6 table and $\omega_\lambda(K_{33})$. $\square$

Theorem 4.10.4 (polynomiality; PROVED — proof obtained separately and checked line by line). For every ρ without parts 1, $K_\rho(n)$ is a polynomial in n and the power sums $\sum_k J_k^m$ ($m \leq |\rho| - \ell(\rho)$) of weighted degree $\leq |\rho| - \ell(\rho)$ (weights $n \mapsto 1, \sum J^m \mapsto m$), valid in $\mathbb{Q}[S_n]$ for all n . Consequently $\omega_\lambda(K_\rho)$ is a polynomial in $n, C_1, \dots, C_{|\rho|-\ell(\rho)}$ of that weighted degree, and each level polynomial $g_i = (n)_i u_i / f^\lambda$ is a polynomial in n and content power sums, invariant under $C_k \mapsto (-1)^k C_k$. The same computation also determines (by full-rank certificates on $n \leq 16$ and exact solving, re-verified there for $n \leq 20$; not independently recomputed) the *new* invariant contributed by each level modulo the previous ones: level 7: $C_6 - 16C_1 C_3$; level 8: $2C_3^2 - (8n + 164)C_1 C_3$; level 9: $C_8 - 24C_1 C_5 - (32n - \frac{7720}{3})C_1 C_3$; level 10: $-24C_1 C_2 C_3 + (12n^2 + 1060n + 62664)C_1 C_3 - 360C_1 C_5$. Every new invariant beyond level 6 involves products of odd content moments.

Separation from the top (COMPUTATIONALLY VERIFIED). $(n, f^\lambda, C_2, C_1^2, C_4)$ — i.e. $(n, d_n, d_{n-3}, d_{n-4}, d_{n-5})$ — separates all transpose classes of partitions of n for every $n \leq 48$ and fails first at $n = 49$, for $(9, 8, 8, 6, 3^5, 1^3)$ vs $(11, 7, 5, 5, 4^4, 3, 1, 1)$ (separate computation to $n \leq 58$; the $n \leq 49$ statement independently re-verified). Adding u_7 separates all $n \leq 58$. With fewer levels: (n, r, f, C_2) fails from $n = 14$, (n, r, f, C_2, C_1^2) from $n = 28$. So the “top separating depth” is unbounded, as is the “bottom separating depth” (Section 4.6).

Proposition 4.10.5 (mirror box moves; PROVED mechanism, existence re-verified). If μ arises from λ by moving a removable box of content $-c$ to an addable position of content $+c$, then $C_k(\mu) = C_k(\lambda)$ for all even k , and $C_1(\mu)^2 = C_1(\lambda)^2$ iff $C_1(\lambda) = -c$. Such pairs with $f^\mu = f^\lambda$ exist, the smallest at $n = 50$: $\lambda = (12, 10, 8, 3, 3, 2^6, 1, 1)$, $\mu = (12, 11, 8, 3, 3, 2^5, 1^3)$ (equal f , equal C_{2k} for all k , $C_1 = \mp 9$; re-verified). Hence **no proof of Conjecture B can use only f^λ and even content moments**; these pairs are separated by $C_1 C_3$ (level 7) and, from the bottom, already by d_2 (their d -vectors agree exactly at $d_n, d_{n-3}, \dots, d_{n-6}$ and differ at d_{n-7} and at $d_2, \dots$; re-verified).

Lemma U and the families F_a (PROVED). If the upper parts $\{\nu \subseteq \lambda : |\nu| \geq s\}$ and $\{\nu \subseteq \mu : |\nu| \geq s\}$ of the lower intervals of Young’s lattice are isomorphic as graded posets, then $d_j(\lambda) = d_j(\mu)$ for $j \leq n - s$. This explains the pairs $F_a = \{(a, 1^{n-a}), (2^{a-1}, 1^{n-2a+2})\}$, $a \geq 3$, which agree on $d_0, \dots, d_{n-2a+2}$; transposing, F_a is the pair $\{(\lambda_1, 1^m), (\lambda_1, m)\}$ of Proposition 4.6.1 with $m = a - 1$. An exhaustive check for $n \leq 40$ shows that for $n \geq 8$ the pair F_3 is the *only* pair of transpose classes agreeing on $d_0, \dots, d_{n-4}$.

Independent review. Six independent adversarial re-derivations (two each) checked Theorem 4.10.1 with its group-algebra identities, Theorem 4.10.4 with the rank-certificate argument, and further results (closed forms for hooks and two-row shapes, sharpness of F_a , Kerov's transition formula), re-deriving every formula with separately written code (identities in $\mathbb{Z}[S_n]$ for $n \leq 9$, Theorem 4.10.1 for all $\lambda \vdash n \leq 24$, etc.). All six confirmed the statements they checked. The only defect found is cosmetic: the proof of the K_3^2 expansion excludes the cycle type $(4, 2)$ by calling it odd, which it is not; it is excluded because a product of two 3-cycles moving six points has disjoint supports, hence type $(3, 3)$.

Further results of the same computation (key data independently re-verified). (a) *Sharpness of F_a* (PROVED): the pair F_a differs by exactly 1 at d_{n-2a+3} — the same statement as Proposition 4.6.1 after transposition, proved there by the two formulas and again, separately, via Aitken's determinant $f^{\lambda/\nu} = |\lambda/\nu|! \det[1/(\lambda_i - \nu_j - i + j)!]$. (b) *Kerov's transition formula* (PROVED, elementary hook-length proof; verified for all $\nu \vdash m \leq 26$): if x is the content of an addable corner of $\nu \vdash m$, X and Y the content sets of the addable and removable corners, then $f^{\nu+x}/f^\nu = (m+1) \prod_{y \in Y} |x-y| / \prod_{x' \in X \setminus x} |x-x'|$; consequently a mirror box move $\lambda \mapsto \mu$ (content $-c \mapsto c$) preserves f iff the two transition ratios of $\lambda' = \lambda - (\text{box } -c)$ towards contents c and $-c$ coincide — an exact but Diophantine criterion. The tempting statement " $f^\mu = f^\lambda \iff C_1(\lambda) = -c$ " is FALSE in both directions (first counterexamples at $n = 20$ and $n = 33$; VERIFIED). (c) *Proposition M0* (PROVED modulo the level polynomials of Theorem 4.10.4): two shapes with the same multiset of absolute contents, $C_1 = 0$ and equal f agree on $d_{n-7}, \dots, d_n$; if also C_3^2 agrees, on $d_{n-10}, \dots, d_n$. (d) *Scan to $n = 84$* (VERIFIED with a C++ sieve plus exact re-checks; the key pair independently re-verified from hook lengths and contents): the number of pairs of transpose classes with equal (n, f, C_2, C_1^2, C_4) is 0 for $n \leq 48$ and grows to 76 at $n = 84$; the **first pair agreeing on the top eight entries $d_{n-7}, \dots, d_n$ occurs at $n = 84$** : $\lambda = (19, 17, 7, 4^6, 3, 2^6, 1, 1)$, $\mu = (19, 16, 8, 8, 3^6, 2^7, 1)$, with equal f (a 56-digit number), $C_1 = 0$, equal C_{2k} for all k , $C_3 = 8298$ vs 1998, $r = 7$ vs 6, and d_{n-8} different (exact skew-tableau counts). So the "top separating depth" is 3 for $n \leq 13$, 4 for $n \leq 27$, 5 for $n \leq 48$, 7 for $n \leq 83$ and ≥ 8 from $n = 84$ on.

4.11 Two-corner shapes (re-verified)

Write a two-corner shape as $\lambda = ((a+c)^b, c^d)$ with corner runs $((a, b), (c, d))$, $a, b, c, d \geq 1$, $n = ab + bc + cd$; transposition is $(a, b, c, d) \mapsto (d, c, b, a)$. The following was obtained in a separate computation; Lemma 4.11.1 was checked by two independent reviewers, and the closed forms and Theorem 4.11.2 were independently re-verified (src/check_agents2.py).

Lemma 4.11.1 (two-rectangle window; PROVED). For $0 \leq j \leq a+d$, $d_j(\lambda) = j! [t^j] G_{a,b}(t) G_{c,d}(t)$, and $d_{a+d+1}(\lambda) = (a+d+1)! [t^{a+d+1}] G_{a,b} G_{c,d} + \binom{a+d}{a}$. *Proof.* A filter of λ meeting the "hinge" rectangle $\{i \leq b, j \leq c\}$ contains at least $a+d+1$ boxes (the rest of its row and column), with equality only for the hook through the box (b, c) , whose removal orders are the $\binom{a+d}{a}$ interleavings of its arm and leg; every smaller filter splits as a filter of the $b \times a$ rectangle plus a filter of the $d \times c$ rectangle with no order relations between the parts (as in Theorem 4.2). $\square$ (This sharpens the window $2m$ of Theorem 4.2 for $r = 2$.)

Theorem 4.11.2 (PROVED). Conjecture B holds for every two-corner shape at least two of whose parameters equal 1 — i.e. for the class $\mathcal{C}$ of Theorem 4.9 together with the families $F_1(c, d) = (c+1, c^d)$, $F_2(b, c) = ((c+1)^b, c)$, $F_4 = F_1^t$ with both parameters ≥ 2 : if $\mu \vdash n$ and $d(\mu) = d(\lambda)$ then $\mu \in \{\lambda, \lambda^t\}$. *Proof sketch (the full case analysis is included with the code).* d_1, d_2 force μ into $\mathcal{C} \cup F_1 \cup F_2 \cup F_4$ (Theorem 4.9, Step 2) and Theorem 4.9 excludes $\mu \in \mathcal{C}$; for the F -families one has closed forms: $d_j(F_2(b, c)) = d_{j+1}((c+1)^{b+1})$ (removing the box $(b+1, c+1)$ first), and with x the smaller parameter, $d_j = I_{j+1}$ for $j \leq x$, $d_{x+1} = I_{x+2} - \varepsilon$

with $\varepsilon = [c = x] + [d = x]$ for F_1 and $1 + [b = c]$ for F_2 , plus explicit d_{x+2} ; comparing x, ε, n and d_{x+2} settles all cases (F_2 vs F_2 reduces to rectangles, Theorem 4.4). All closed forms re-verified here for all such shapes with $n \leq 44$, and the theorem itself against all partitions of $n \leq 26$.

Conjecture D (COMPUTATIONALLY VERIFIED for $n \leq 300$, 604,513 shapes). (n, f^λ, C_2) determines a two-corner shape up to transpose. Since all three are determined by the d -vector, Conjecture D would give Conjecture B for all two-corner shapes; it is a purely Diophantine statement about the hook product and the cubic polynomial $C_2(a, b, c, d)$. Neither invariant suffices alone: (n, C_2) collides already at $n = 9$, and even shapes with the same multiset $\{a, b, c, d\}$ can share (n, C_2) ((6, 7, 8, 10) vs (7, 10, 6, 8), $n = 178$). The bottom of the vector is provably local (it reads only the parameters below $x_1 + x_2$ where $x_1 \leq x_2$ are the two smallest), so the general two-corner case must use the global invariants.

4.12 Structural results (re-verified)

The following results were obtained in a separate computation; the character-sum formula is proved here directly from Lemma 4.5, and the two row-bounded theorems were checked line by line and their sign thresholds independently re-verified (src/check_agents2.py).

Theorem 4.12.1 (character sum over square roots; PROVED). For $\lambda \vdash n$,

$$n! P_\lambda(t) = \sum_{\tau \in S_n} \chi^\lambda(\tau^2) (1+t)^{\text{fix}(\tau)}, \quad d_j(\lambda) = \frac{j!}{n!} \sum_{\tau \in S_n} \chi^\lambda(\tau^2) \binom{\text{fix}(\tau)}{j}.$$

Proof. By Lemma 4.5 and Frobenius-Schur, $u_i = \sum_{\rho \vdash i} \frac{\sigma(\rho)}{z_\rho} \chi^\lambda(\rho \cup 1^{n-i}) = \frac{1}{i!} \sum_{h \in S_i} \chi^\lambda(h^2 \cup 1^{n-i})$. On the other hand $\sum_{\tau \in S_n} \chi^\lambda(\tau^2) (1+t)^{\text{fix}(\tau)} = \sum_{A \subseteq [n]} t^{|A|} \sum_{\tau \text{ fixing } A \text{ pointwise}} \chi^\lambda(\tau^2) = \sum_j \binom{n}{j} t^j \sum_{h \in S_{n-j}} \chi^\lambda(h^2 \cup 1^j) = \sum_j \binom{n}{j} t^j (n-j)! u_{n-j} = n! \sum_j d_j \frac{t^j}{j!}$. $\square$ (Verified by brute force over S_n for $n \leq 7$.) Consequently $n! P_\lambda(-1) = \sum_{\tau \text{ derangement}} \chi^\lambda(\tau^2)$, and the d -vector is equivalent to the integer vector $(\sum_{\tau \in \text{Der}_m} \chi^\lambda(\tau^2 \cup 1^{n-m}))_{m \leq n}$ of “derangement transforms”.

Theorem 4.12.2 (at most three rows; PROVED). If $\lambda, \mu \vdash n$ both have at most 3 rows and $d(\lambda) = d(\mu)$, then $\mu \in \{\lambda, \lambda^t\}$; for $n \geq 10$, $\mu = \lambda$. *Proof.* Put $x_i = \lambda_i - i$ ($i = 1, 2, 3$, zero-padded), $p_k = \sum x_i^k$. Summing the contents row by row, $n = p_1 + 6$, $C_1 = \frac{p_2 + p_1}{2} - 4$, $C_2 = \frac{2p_3 + 3p_2 + p_1}{6} + 6$. Row i of length ℓ contributes $\ell(\ell+1-2i)/2$ to C_1 , so $2C_1 \geq \lambda_1(\lambda_1-1) - 8$, and $C_1 \leq 0$ forces $\lambda_1 \leq 3$, $n \leq 9$ (the largest such shape is (3, 3, 3); re-verified: no ≤ 3 -row shape with $n \geq 10$ has $C_1 \leq 0$). For $n \geq 10$ the d -vector determines C_1^2 and C_2 (Cor. 4.10.2), hence $C_1 > 0$ itself, hence p_1, p_2, p_3 , hence by Newton’s identities the monic cubic $\prod (T - x_i)$, hence the strictly decreasing triple (x_i) and λ . For $n \leq 9$ use Theorem 4.1. $\square$

Theorem 4.12.3 (at most four rows; PROVED). If $\lambda, \mu \vdash n$ both have at most 4 rows, $n \geq 17$, and $d(\lambda) = d(\mu)$, then $\mu = \lambda$ (and $\mu \in \{\lambda, \lambda^t\}$ for all n , by Theorem 4.1 below 17). *Proof.* With four coordinates $x_i = \lambda_i - i$: $n = p_1 + 10$, $C_1 = \frac{p_2 + p_1}{2} - 10$, $C_2 = \frac{2p_3 + 3p_2 + p_1}{6} + 20$, $C_4 = \frac{6p_5 + 15p_4 + 10p_3 - p_1}{30} + 116$. The sign lemma gives $2C_1 \geq \lambda_1(\lambda_1-1) - 20$, so $C_1 > 0$ for $n \geq 21$; an exhaustive check of the ≤ 4 -row shapes with $n \leq 20$ shows that the largest one with $C_1 \leq 0$ is (4, 4, 4, 4), $n = 16$, so $C_1 > 0$ for all $n \geq 17$ (the original derivation stated $n \geq 21$; the exhaustive check gives 17). Thus p_1, p_2, p_3 and e_1, e_2, e_3 are determined; Newton’s identities $p_4 = e_1 p_3 - e_2 p_2 + e_3 p_1 - 4e_4$ and $p_5 = e_1 p_4 - e_2 p_3 + e_3 p_2 - e_4 p_1$ make $30(C_4 - 116)$ an affine function of e_4 with slope $-30(e_1 + 2) = -30(n - 8) \neq 0$; so e_4 , the quartic $\prod (T - x_i)$, and λ are determined. $\square$ *Remarks.* Both theorems are within-class statements (both shapes

row-bounded); five rows would need one more even-degree invariant independent of C_1^2, C_2, C_4 (the level-7 invariant $C_6 - 16C_1C_3$ involves an odd moment).

Theorem 4.12.4 (Pfaffian expansion; PROVED separately, three-row corollary independently re-verified for all three-row shapes with $n \leq 18$). Expanding the Pfaffian of Theorem 4.7 along its last index (with $\ell > \ell(\lambda)$ even, so the last entries are truncated exponentials $E_k(t) = \sum_{j \leq k} t^j/j!$) gives $P_\lambda = \sum_{j < \ell} (-1)^{j+1} E_{\alpha_j-1} \text{Pf}(B^{(j\ell)})$; for three rows,

$$P_{(a,b,c)}(t) = E_c(t)P_{(a+2,b+2)}(t) - E_{b+1}(t)P_{(a+2,c+1)}(t) + E_{a+2}(t)P_{(b+1,c+1)}(t),$$

with the two-row polynomials of Corollary 4.8. (A negative result from the same computation, VERIFIED for $n \leq 10$: there is no constant-coefficient bilinear differential identity expressing $\sum_c P_{\lambda-c}(t)P_{\lambda-c}(u)$ through P_λ and its derivatives, so the multiset of children polynomials cannot be read off P_λ by such a formula.)

4.13 Attempts to disprove Conjecture B: independent confirmation, a both-ends family, and a heuristic

A separate effort, whose proved claims were checked by two independent reviewers, tried to disprove Conjecture B.

Independent confirmation (COMPUTATIONALLY VERIFIED). From-scratch d -vector code reproduces, for every $n \leq 42$, the counts, the number of transpose pairs, the absence of collisions and the digests of `data/collisions_python.txt`. A targeted search for pairs of transpose classes with the same multiset of corner boxes, the same f^λ and the same (C_2, C_1^2, C_4) finds none for $n \leq 52$.

Theorem 4.13.1 (a family agreeing at both ends; PROVED separately, independently re-verified for $k \leq 7$). For $k \geq 1$ let $\lambda_k = (5, 3^k, 2, 2, 2)$ and $\mu_k = (4, 4, 3^k, 1, 1, 1)$, $n = 3k + 11$. Then $f^{\lambda_k} = f^{\mu_k}$, $C_2(\lambda_k) = C_2(\mu_k)$, $C_1(\lambda_k) - C_1(\mu_k) = 2$, and $d_j(\lambda_k) = d_j(\mu_k)$ for $0 \leq j \leq k$ and for $n - 3 \leq j \leq n$, while d_{n-4} differs. So the two shapes agree on a prefix of length about $n/3$ and on the top four entries (the first place where C_1^2 enters, d_{n-4} , separates them). The bottom agreement rests on a “body insertion” lemma reducing it to $d_j((3^K, 2, 2, 2)) = d_j((3^K, 1, 1, 1))$ for $j \leq K$, which after transposition and a 180° rotation becomes a symmetry $N_j(\nu) = N_j(\nu^*)$ of chain counts in the three-row part of Young’s lattice, $\nu^* = (\nu_1 - \nu_3, \nu_1 - \nu_2, 0)$; this is proved via the Racah-Speiser form of Weyl’s character formula for GL_3 (the number of irreducible constituents of $V_\lambda \otimes W$ equals that of $V_\lambda \otimes W^*$), reduced to a sign identity $\sum_{\sigma \in S_3} \text{sgn}(a + \sigma b) = \sum_{\sigma} \text{sgn}(a - \sigma b)$ that is piecewise constant on the cells of a hyperplane arrangement and is settled by a finite check. (Direct computation for $k \leq 7$ shows agreement even on $d_0, \dots, d_{k+3}$; the exact disagreement window is $\{k+4, \dots, n-4\}$ for $k \leq 12$ by the separate computation.) Together with Proposition 4.6.1 and Proposition 4.10.5 this shows that bounded-length agreement at either end, or at both ends simultaneously with a linearly growing middle window, is compatible with non-conjugacy; **a counterexample would have to be controlled in the middle as well.**

Heuristic (CONJECTURAL, honest version). Among the 26,613 transpose classes at $n = 42$ the conditional probability that two classes agreeing on $d_0, \dots, d_{a-1}$ also agree on d_a rises with a (to about 0.75 at $a = 15$): the bottom of the vector is “sticky”, but every deeply agreeing pair found is structural (attachment or body-insertion pairs) and provably separates at the top. From the top, $P(\text{same } f) \approx 1.4 \cdot 10^{-5}$, $P(\text{same } d_{n-3} \mid \text{same } f) \approx 0.017$, $P(\text{same } d_{n-4} \mid \dots) \approx 0.012$, and no pair with $n \leq 42$ agrees on the top six entries. If successive top levels behaved like independent polynomial invariants with conditional agreement probability $\leq c < 1$, the expected number of collisions at size n would be at most $T_5(n) c^{n/2-6}$ with $T_5(n)$ (the number of depth-5 collisions, Section 4.10) growing only slowly — negligible for $n \geq 60$ if $c \leq 0.1$. The only

real input of this heuristic is genericity of the top levels relative to one another; every known mechanism forcing agreement (mirror box moves, body insertion, attachment) has bounded reach at one end and provably fails at the other. So a counterexample cannot be an accident of small numbers; it would need an identity controlling all $\approx n - 4$ coordinates at once, as transposition does. The verdict: unlikely but not excluded.

5. The shifted tree (Problem 2)

Setting. A strict partition $\lambda = (\lambda_1 > \dots > \lambda_\ell > 0)$ has shifted diagram with row i occupying the boxes $(i, i), \dots, (i, i + \lambda_i - 1)$. Covers in the shifted Young lattice: a box can be added to row i iff $i = 1$ or $\lambda_{i-1} \geq \lambda_i + 2$, and a new row of length 1 can be started iff $\lambda_\ell \geq 2$ (or $\lambda = \emptyset$); a box can be removed from row i iff $i = \ell$ or $\lambda_i \geq \lambda_{i+1} + 2$. The tree SSYT of shifted standard tableaux (= saturated chains from $\emptyset$), its census $s_k(\lambda)$ and down-census $d_j(\lambda)$ are defined as before; g^λ is the number of shifted SYT of shape λ (shifted hook formula $g^\lambda = \frac{n!}{\prod \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}$). Worley's Problem 2 conjectures that SSYT has only the trivial automorphism and that each node is determined by its census (necessarily: among nodes of the same depth, since all tableaux of one shape have identical subtrees). The computations in this section (src/shifted.py, src/shifted_scan.py, tests/test_shifted.py) were independently re-derived with a separately written box-set model of shifted diagrams (cover sets identical for all strict partitions of $n \leq 21$; all census data recomputed and identical for $n \leq 40$, $K = 14$); the structural claims below were re-checked by hand.

5.1 (PROVED.) Basic structure. (a) $\# \text{addable}(\lambda) = \# \text{removable}(\lambda) + 1 - [\lambda_\ell = 1]$ for $\lambda \neq \emptyset$; hence with U the (unweighted) up operator and D the unweighted down operator, $DU - UD = I - \Pi$ where Π is the projection onto strict partitions with last part 1 (all off-diagonal coefficients cancel exactly as in Lemma 2.2), while with D' the down operator weighted by 2 for removing an off-diagonal box and 1 for a diagonal box, $D'U - UD' = I$ (Fomin's dual graded graphs; verified on all strict partitions of $n \leq 10$, data/shifted_fomin.txt). (b) λ has exactly one addable box iff $\lambda = \emptyset$ or λ is a staircase $(\ell, \ell - 1, \dots, 1)$; every strict partition has an addable box (row 1). Hence the root is the unique vertex of degree 1 (staircase vertices have degree 2, all others ≥ 3), every automorphism fixes the root and preserves depth, and — exactly as in Theorem 3.7 — *sibling rigidity* (distinct children of a vertex have non-isomorphic subtrees; here children have distinct shapes, so it suffices that distinct strict partitions of the same size have different censuses) for all sizes implies $\text{Aut}(\text{SSYT}) = \{1\}$, by induction on depth.

5.2 (PROVED.) There is no shifted analogue of Theorem 2.3 with a universal factor. Suppose $E_\lambda(t) = \Phi(t) Q_\lambda(t)$ for all strict λ , with Φ universal and Q_λ a polynomial of degree $\leq |\lambda|$ (e.g. the EGF of any down-census, weighted or not). Since $\emptyset$ has the unique cover (1) , $s_k((1)) = s_{k+1}(\emptyset)$, i.e. $E_{(1)} = E'_\emptyset$; and $Q_\emptyset$ is a constant, so $Q_{(1)}/Q_\emptyset = E_{(1)}/E_\emptyset = (\log E_\emptyset)'$ would be a polynomial of degree ≤ 1 . But $E_\emptyset(t) = \sum_k (\# \text{shifted SYT of size } k) t^k/k! = 1 + t + \frac{t^2}{2} + \frac{2t^3}{6} + \frac{3t^4}{24} + \dots$ (1, 1, 1, 2, 3, 6, 12, 27, 63, 154, 398, ...), so $\log E_\emptyset = t + \frac{t^3}{6} - \frac{t^4}{12} + \dots$ and $(\log E_\emptyset)' = 1 + \frac{t^2}{2} - \frac{t^3}{3} + \dots$ has a nonzero t^2 coefficient. (Remark, using standard saddle-point asymptotics: even allowing Q_λ of arbitrary degree, $(\log E_\emptyset)'$ is not a polynomial. Indeed $E_\emptyset = e^{q(t)}$ with q a polynomial of degree D forces the number a_k of shifted SYT of size k to satisfy $a_k^{1/k} \asymp k^{1-1/D}$, whereas $a_k \leq \prod_{j < k} (\sqrt{2j} + 2) \leq (\sqrt{2k} + 2)^k$ because a strict partition of j has at most $\sqrt{2j}$ parts and hence at most $\sqrt{2j} + 1$ addable boxes; so $D \leq 2$, and the t^3 -coefficient $\frac{1}{6}$ of $\log E_\emptyset$ rules out $D \leq 2$.) In the unshifted case $(\log E_\emptyset)' = 1 + t = P_{(1)}(t)$, consistent with Theorem 2.3. What does hold (verified for all strict $|\lambda| \leq 8$, $k < 10$) is the corrected recursion $A'_\lambda = \frac{1}{2} A_{D'\lambda} + (1 + \frac{t}{2}) A_\lambda - \frac{1}{2} C_\lambda$, where C_λ is the EGF of the number

of upward chains of length k from λ ending at a partition with last part 1; the extra term C_λ is what prevents a product formula. Consequently the shifted census cannot be reduced to a finite vector by this route, and the census conjecture was tested through truncations.

5.3 (COMPUTATIONALLY VERIFIED.) Census injectivity and sibling rigidity. Let $q(n)$ be the number of strict partitions of n . The truncated census $(s_0, \dots, s_K)(\lambda)$ was computed by level recursion for all strict $\lambda \vdash n$: for $n \leq 40$ with $K = 14$ and for $n \leq 60$ with $K = 21$ (data/shifted_census_N40_K14.txt, data/shifted_census_N60_K21.txt). **For every $n \leq 60$ the $q(n)$ strict partitions of n have pairwise distinct truncated censuses** (e.g. $q(40) = 1113$, $q(60) = 10880$). Since the truncated census is a function of the full census, the full census determines the shape among strict partitions of the same size for all $n \leq 60$; sibling rigidity therefore holds for all vertices of depth ≤ 59 (also checked directly: 43,806 sibling pairs at depths ≤ 39 with $K = 14$, none equal), and **every automorphism of SSYT fixes every vertex of depth ≤ 60** . The smallest separating depth K_n grows roughly like $n/3$ ($K_{40} = 13$, $K_{60} = 19$; the hardest pairs are two-row shapes such as $(28, 12)$ vs $(26, 14)$), so no bounded-depth invariant can work for all n , and — unlike the unshifted case — no finite reformulation is available. A fixed truncation does not determine n across sizes (e.g. $(15), (16), \dots, (40)$ share $(s_0, \dots, s_{14})$), which is irrelevant for automorphisms (depth is preserved).

5.4 (CONJECTURAL.) Worley’s Problem 2 (both halves) remains open beyond depth 60.

6. Open sub-questions

1. **Conjecture B** in general (CONJECTURAL; verified $n \leq 75$). Equivalent finite form: the vector $(d_j(\lambda))_{j \leq n}$ determines λ up to transpose.
2. **B''** (CONJECTURAL; verified for $|\lambda| \leq 74$): would already give the full description of $\text{Aut}(\text{SYT})$ (Theorem 3.7). It follows from the special case of B for two shapes differing by moving one box. It is *not* reducible to simple invariants: non-mirror siblings can share f (e.g. $(5, 2)$ and $(4, 3)$, children of $(4, 2)$, both with $f = 14$), (f, r) , (r, d_2, f) , and even (f, C_2) (two sibling pairs with $n \leq 34$, e.g. the children $(11, 7, 3, 3, 2, 2, 1^4)$ and $(11, 6, 3, 3, 2, 2, 2, 1^3)$ of $(11, 6, 3, 3, 2, 2, 1^4)$; src/explore_siblings.py, data/siblings_invariants_n34); while the full d -vector separates all 719,562 non-mirror sibling pairs with $n \leq 34$. Is B'' strictly weaker than B'? Is B' strictly weaker than B? (No examples known either way.)
3. Which content polynomials beyond C_2, C_1^2, C_4 (Cor. 4.10.2) and the level-7-10 invariants of Theorem 4.10.4 are determined by the d -vector; note that even content moments together with f^λ can never suffice (Prop. 4.10.5), so odd-moment products or the bottom of the vector are essential.
4. Does P_λ determine the multiset $\{P_{\lambda-c}\}_c$ of its “children” polynomials (only $\sum_c P_{\lambda-c} = P'_\lambda$ is obvious)? A positive answer would give an inductive proof of B.
5. **Conjecture D** (Section 4.11): (n, f^λ, C_2) determines a two-corner shape up to transpose (verified to $n = 300$); it would settle B for all two-corner shapes. Extending Theorems 4.12.2–4.12.3 to five rows needs one more even-degree content invariant that is provably determined by the d -vector; and a version across row-classes needs the number of rows (or columns) to be recoverable.
6. The shifted analogue: an identity $E_\lambda^{\text{sh}} = \Phi \cdot P_\lambda^{\text{sh}}$ with a universal Φ (see Section 5).

7. Assessment

Problem 3 has an automorphism half and a census half. The automorphism half is, in our judgement, settled up to one lemma: Worley’s conjectured generators are provably insufficient (Theorem 3.5 — a small but genuine correction to the conjecture as stated), the corrected

generators (pair transposes) are the right ones if and only if the sibling-rigidity statement B'' holds (Theorem 3.7), and B'' follows from the census conjecture B , which is verified for all shapes of size ≤ 75 (so the description of $\text{Aut}(\text{SYT})$ is unconditional to depth 75). B'' only concerns two shapes differing in a single box and may well be more accessible than B itself; proving it would completely answer the automorphism question. The census half (B) is not solved. We have the finite reformulation (Theorem 2.3, which turned out to be a 2003 theorem of Stanley), verification to $n = 75$, proofs for rectangles, for all shapes avoiding $(3, 2, 1)$, for two-corner shapes with two unit parameters, and within the classes of shapes with at most three or four rows (Theorems 4.4, 4.9, 4.11.2, 4.12.2–4.12.3), a list of invariants that the d -vector provably determines (Theorems 4.2, 4.6), and a proved infinite family of near-collisions (Proposition 4.6.1) showing that any proof must use the very top of the vector — for $n \leq 45$ the coordinate d_{n-3} is in fact always what finally separates shapes. From the top, the first five informative entries (f, C_2, C_1^2, C_4) separate everything up to $n = 48$ but not beyond; pairs with equal f and equal even content moments exist from $n = 50$ on, and at $n = 84$ a pair agrees on the top eight entries (Section 4.10). We found no structural mechanism (the shifted-symmetric-function count in Section 4.7 shows that the obvious “generic” route cannot work), so a proof of B probably needs a new idea, e.g. an interpretation of the coefficients of P_λ beyond $j = 2m$ that sees the order and orientation of the corners; and a counterexample at some large n cannot be excluded; the verification range, the targeted searches to $n = 52$, and the heuristic of Section 4.13 make one seem unlikely, but the proved both-ends family (Theorem 4.13.1) shows that agreement of long prefixes and of the whole top is not by itself a contradiction. What a human mathematician should check first: (i) in Theorem 3.5, that $E_{\lambda,p}$ is well defined on Aut_1 and that its value on τ_{T_0} is f^{λ/λ_0} independently of p ; (ii) the uniqueness/freeness part of Theorem 3.7 and the topology used to make “infinite composition” precise; (iii) in Theorem 4.2, step (a) (adjacency generates the induced order on a filter) and step (d) (components stay inside their corner regions); (iv) the trace computation with Jucys–Murphy elements in Theorem 4.6; (v) the case analysis of Theorem 4.9, every formula of which was machine-checked and which was independently re-derived by a second route; (vi) that the mod- p scan is a valid negative test (it is: an exact collision persists modulo every prime), and that the C++ ranking of partitions is correct (it agrees with the exact Python scan on every level $n \leq 60$). For Problem 2 (shifted), the situation is weaker: there is provably no universal-factor identity (Section 5.2), so the census cannot be reduced to a finite vector by the same route, and both halves are verified only to depth 60.

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